SCIENCE CHINA PRESS

SCIENCE CHINA Life Sciences
life.scichina.com link.springer.com
https://doi.org/10.1007/s11427-024-2736-7

REVIEW

Sequencing and characterizing human mitochondrial genomes in the biobank-based genomic research paradigm

Lintao Luo[1, 2 †], Mengge Wang[1, 3, 4 †, *], Yunhui Liu[1, 2 †], Jianbo Li[2], Fengxiao Bu[1, 3], Huijun Yuan[1, 3 *], Renkuan Tang[2 *], Chao Liu[2, 4 *] & Guanglin He[1, 3, 4 † *]

[1]Institute of Rare Diseases, West China Hospital of Sichuan University, Sichuan University, Chengdu 610000, China
[2]Department of Forensic Medicine, College of Basic Medicine, Chongqing Medical University, Chongqing 400331, China
[3]Center for Archaeological Science, Sichuan University, Chengdu 610000, China
[4]Anti-Drug Technology Center of Guangdong Province, Guangzhou 510230, China

†Contributed equally to this work
*Corresponding authors (Mengge Wang, email: Menggewang2021@163.com; Huijun Yuan, email: yuanhj301@163.com; Renkuan Tang, email: Renktang2012@163.com; Chao Liu, email: liuchaogzf@163.com; Guanglin He, email: guanglinhescu@163.com)

Received 25 April 2024; Accepted 18 September 2024; Published online 21 January 2025

Human mitochondrial DNA (mtDNA) harbors essential mutations linked to aging, neurodegenerative diseases, and complex muscle disorders. Due to its uniparental and haploid inheritance, mtDNA captures matrilineal evolutionary trajectories, playing a crucial role in population and medical genetics. However, critical questions about the genomic diversity patterns, inheritance models, and evolutionary and medical functions of mtDNA remain unresolved or underexplored, particularly in the transition from traditional genotyping to large-scale genomic analyses. This review summarizes recent advancements in data-driven genomic research and technological innovations that address these questions and clarify the biological impact of nuclear-mitochondrial segments (NUMTs) and mtDNA variants on human health, disease, and evolution. We propose a streamlined pipeline to comprehensively identify mtDNA and NUMT genomic diversity using advanced sequencing and computational technologies. Haplotype-resolved mtDNA sequencing and assembly can distinguish authentic mtDNA variants from NUMTs, reduce diagnostic inaccuracies, and provide clearer insights into heteroplasmy patterns and the authenticity of paternal inheritance. This review emphasizes the need for integrative multi-omics approaches and emerging long-read sequencing technologies to gain new insights into mutation mechanisms, the influence of heteroplasmy and paternal inheritance on mtDNA diversity and disease susceptibility, and the detailed functions of NUMTs.

mitochondrial DNA | heteroplasmy | NUMT | evolutionary medicine | genomic database

INTRODUCTION

Mitochondrial DNA (mtDNA) is crucial for understanding cellular energy metabolism, numerous human diseases, and evolutionary dynamics. Unlike nuclear DNA (nDNA), mtDNA is inherited maternally, and has a higher mutation rate, and serves distinct functions, offering unique insights into fundamental biological processes. Although the structure of mtDNA is simple (Figure 1A and B), it encodes key components of the oxidative phosphorylation (OXPHOS) system, which are essential for ATP production. The 14 protein-coding genes in mtDNA are vital for OXPHOS complex assembly and function, underscoring the crucial role of mtDNA in metabolic energy conversion (Anderson et al., 1981; Hu et al., 2024; Wallace, 1999) (Figure 1A). Mutations in these genes can cause mitochondrial dysfunction, leading to reduced ATP production and increased reactive oxygen species. These changes are characteristic of various metabolic and degenerative conditions (Schapira, 2006). The significance of mtDNA in disease research is extensive, encompassing mitochondrial disorders caused by direct mtDNA mutations and complex diseases involving indirect mitochondrial dysfunction (Taylor and Turnbull, 2005). Mitochondrial diseases (MDs) display diverse symptoms and inheritance patterns, emphasizing the role of mtDNA in cellular health. Heteroplasmy, the presence of multiple mtDNA haplotypes within a cell, adds complexity to disease phenotypes and inheritance, presenting a rich area for genetic and clinical research (Carelli et al., 2004). MtDNA is also a powerful evolutionary tool for studying human origins and migration patterns due to its maternal inheritance, lack of recombination and high copy number (Figure 1C and D). Its high mutation rate acts as a molecular clock (Figure 1C), facilitating the reconstruction of phylogenetic relationships and human demographic histories (Forster et al., 1996; Pakendorf and Stoneking, 2005). Through mtDNA analysis, researchers have traced human origins, migrations, and patterns of population divergence and admixture, enhancing our understanding of global evolutionary dynamics. The advent of biobanks and advancements in sequencing technologies have

Citation: Luo, L., Wang, M., Liu, Y., Li, J., Bu, F., Yuan, H., Tang, R., Liu, C., and He, G. (2025). Sequencing and characterizing human mitochondrial genomes in the biobank-based genomic research paradigm. Sci China Life Sci 68, 1610–1625. https://doi.org/10.1007/s11427-024-2736-7

CrossMark click for updates

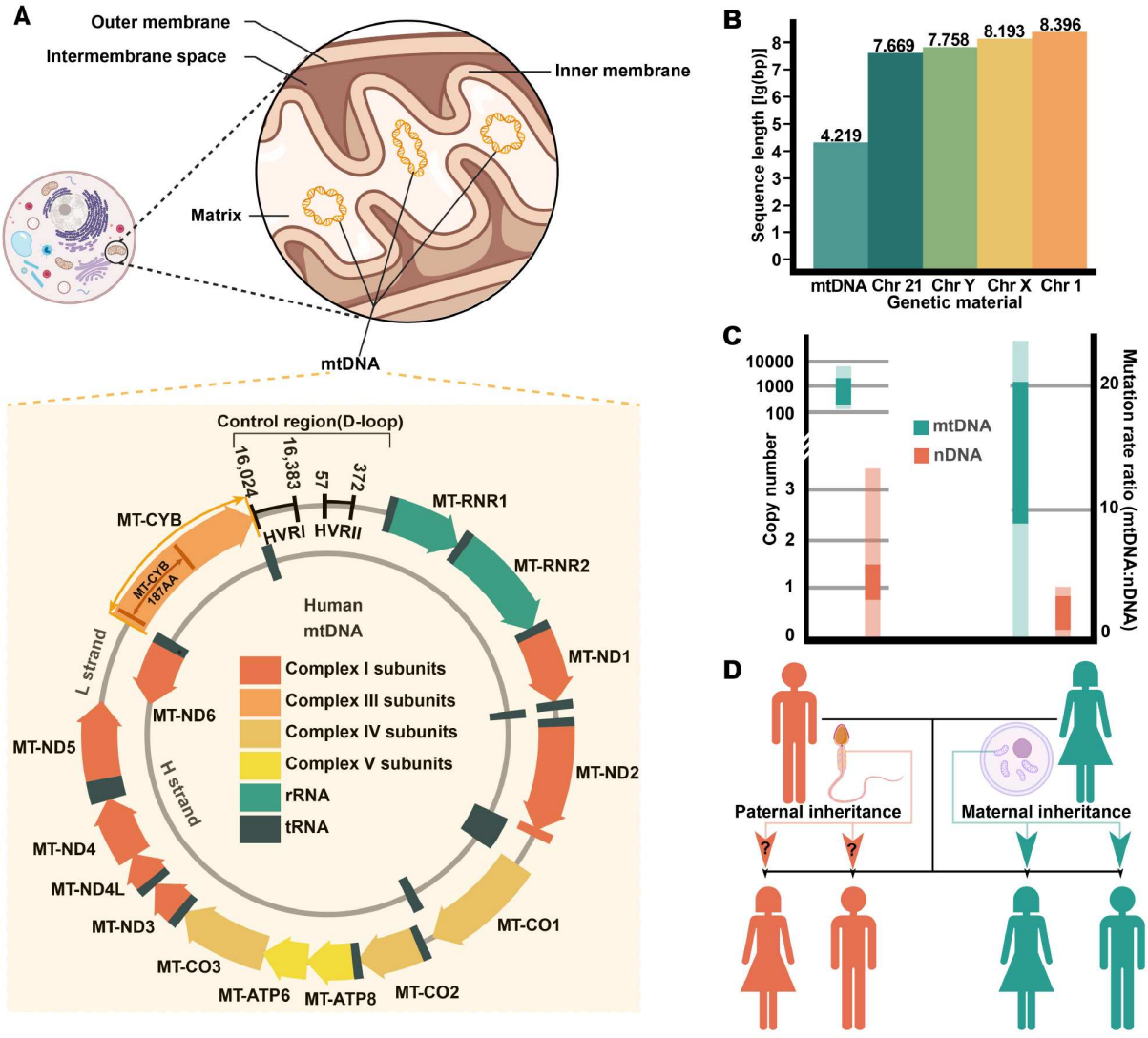


Figure 1. Structure and genetic features of mitochondria and mtDNA. A, The upper left section depicts a human cell, with an enlarged mitochondrion shown on the upper right. The lower section presents an enlarged image of mtDNA, illustrating its structure and encoded products in various colors. Arrows indicate the direction of synthesis. B, A histogram compares the length of mtDNA with that of the longest autosome (Chr 1), the shortest autosome (Chr 21), and the sex chromosomes (Chr Y and Chr X) in a human cell. Sequence lengths are log-transformed, highlighting that mtDNA length is significantly shorter than that of other chromosomes. C, Comparison of copy number and mutation rate between mtDNA and nuclear DNA (nDNA) in a single human cell. The mutation rate ratio (mtDNA: nDNA) is based on the average number of mutations per generation, with slight variations among individuals and populations (Howell et al., 2003; Wallace et al., 1987). Despite differences across various tissues, such as the liver, heart, and brain, mtDNA generally exhibits a greater copy number and mutation rate than nDNA (data source: https://www.genome.ucsc.edu). D, Inheritance pattern of mtDNA. MtDNA is maternally inherited, originating solely from the mitochondria in the mother's egg cells, as it cannot recombine (Giles et al., 1980). The possibility of paternal inheritance of mtDNA, involving transmission through the father's mitochondria, remains debated, represented by a question mark on the arrow to reflect this uncertainty.

revolutionized mitochondrial research. Biobanks preserve high-quality biological samples, providing essential genetic material for comprehensive mtDNA studies. Concurrently, next- and third-generation sequencing (NGS and TGS) technologies enable rapid, high-resolution sequencing and analysis of mtDNA, allowing for more precise characterization of its variants across populations and diseases (Mardis, 2008; Rehm, 2017). This synergy between biobanks and advanced sequencing technologies marks a paradigm shift in genomic research, enhancing our understanding of the role of mtDNA in health, disease, and human history (Figure 2).

This review summarizes the recent advances in the evidence for paternal and maternal inheritance models, the complex genomic landscape of mtDNA heteroplasmy, its distribution patterns in genetically diverse populations, and its implications for understanding human diseases and evolutionary genetics. The influence of mtDNA variations on genetic disorder susceptibility and human adaptive evolution is examined. Additionally, the biological and genetic implications of nuclear-mitochondrial segments (NUMTs) and their controversial role in paternal transmission are discussed. Advances in sequencing technologies have enhanced the accuracy of identifying and categorizing mtDNA heteroplasmy and NUMTs. The role of mtDNA in unraveling human evolutionary history and migration patterns is highlighted, utilizing interdisciplinary research strategies, including computational biology and statistical genetics. In the era of NGS and TGS, we assess the impact of these methods on mtDNA research and the application of mitogenome data in precision medicine, along with

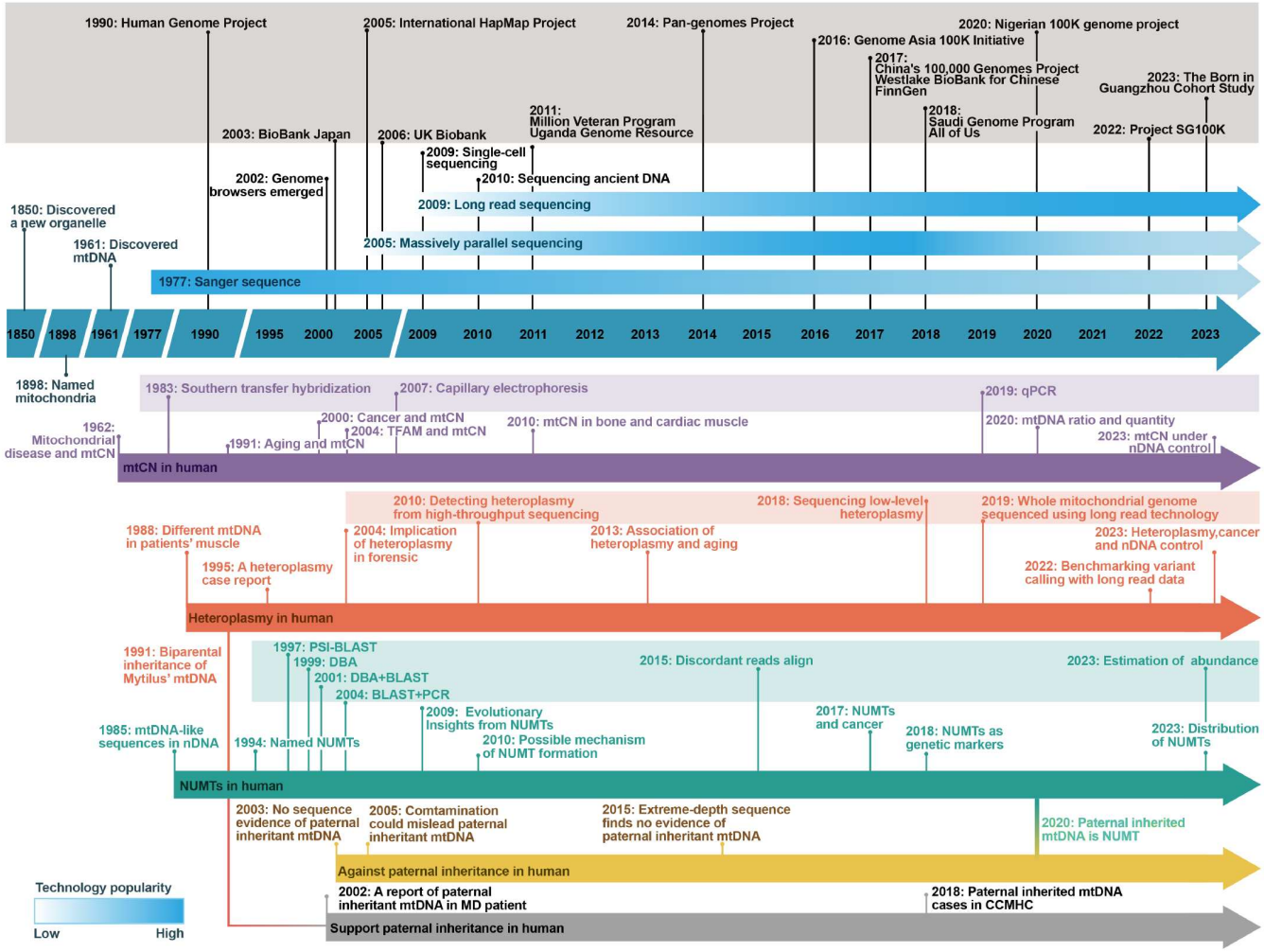


Figure 2. Timeline of mtDNA research and technological advances. The light gray background highlights key human genomic projects, each addressing various scientific questions and providing invaluable insights related to mtDNA research. The blue timeline represents the development of sequencing technologies, with the intensity of the color indicating their popularity. Purple arrows denote research on human mtCN (mitochondrial DNA copy number). Initially, an increase in mtCN was linked to mitochondrial diseases (Luft et al., 1962). Subsequent research indicated that aging and cancer occurrence are also associated with elevated mtCN (Sugiyama et al., 1991). Further studies suggested that the absolute mtCN is more crucial than the normal/abnormal ratio of mtDNA (Filograna et al., 2021). mtCN varies among different tissues and is regulated by specific nuclear genes (Ekstrand, 2004). Recent studies have identified additional nuclear genes that regulate mtCN while also questioning the previously suggested relevance of mtCN to diseases and aging (Gupta et al., 2023). Jacinth arrows reflect research on human heteroplasmy. Early on, heteroplasmy in humans became widespread, with pathogenic variants potentially related to aging and cancer (Comas et al., 1995; Wallace et al., 1988). A challenging question is the origin of different mtDNA types, whether they are *de novo* mutations or inherited from parents. Research on Mytilus suggested that offspring mtDNA types include mixtures of maternal and paternal mtDNA types, suggesting the theory of paternal inheritance of mtDNA (gray axis) (Hoeh et al., 1991). Green arrows represent research on human NUMTs (nuclear mitochondrial DNA segments). NUMTs were discovered serendipitously when researchers found sequences highly similar to mtDNA during the alignment of nuclear DNA, leading to the hypothesis that mtDNA fragments might be inserted into the nuclear genome (du Buy and Riley, 1967; Nomiyama et al., 1985). Yellow and gray arrows reflect the debate over the mode of human mtDNA inheritance. Although some case reports support paternal inheritance (Luo et al., 2018; Schwatz and Vissing, 2002), more studies have not found evidence of paternal inheritance (Bandelt, 2005; Filosto et al., 2003; Schwartz and Vissing, 2004). Some research suggests that "paternal mtDNA" consists of NUMTs (Wei et al., 2020; Wei et al., 2022). Lighter shades in the background color illustrate the development of corresponding technologies. The gray and yellow arrows present viewpoints supporting or opposing the paternal inheritance of human mtDNA. Significant events and corresponding time points are marked in their respective colors above the axes.

associated ethical and technological challenges. The key findings and contributions of this work are summarized, with an emphasis on the role of biobanks in advancing mitogenome research. Recommendations for future research directions are provided, focusing on current challenges and trends, particularly in harnessing resources and new technologies.

Landscape of mtDNA heteroplasmy

Heteroplasmy is the presence of more than one type of mtDNA sequence within a single cell or organism, which can be further categorized into length heteroplasmy and sequence heteroplasmy (He et al., 2010) (Figure 3A and B). Length heteroplasmy results from insertions/deletions (InDels) or duplications altering the length of the mtDNA molecule. Sequence heteroplasmy involves nucleotide sequence variations within the same individual, primarily single nucleotide variants (SNVs) (Samuels et al., 2013) (Figure 3A). DNA mutations causing heteroplasmy can stem from internal factors, such as DNA replication mismatches, or external factors, such as ultraviolet light, radioactive radiation, chemicals, and certain viruses. Previous studies have shown correlations between age and heteroplasmy (Michikawa

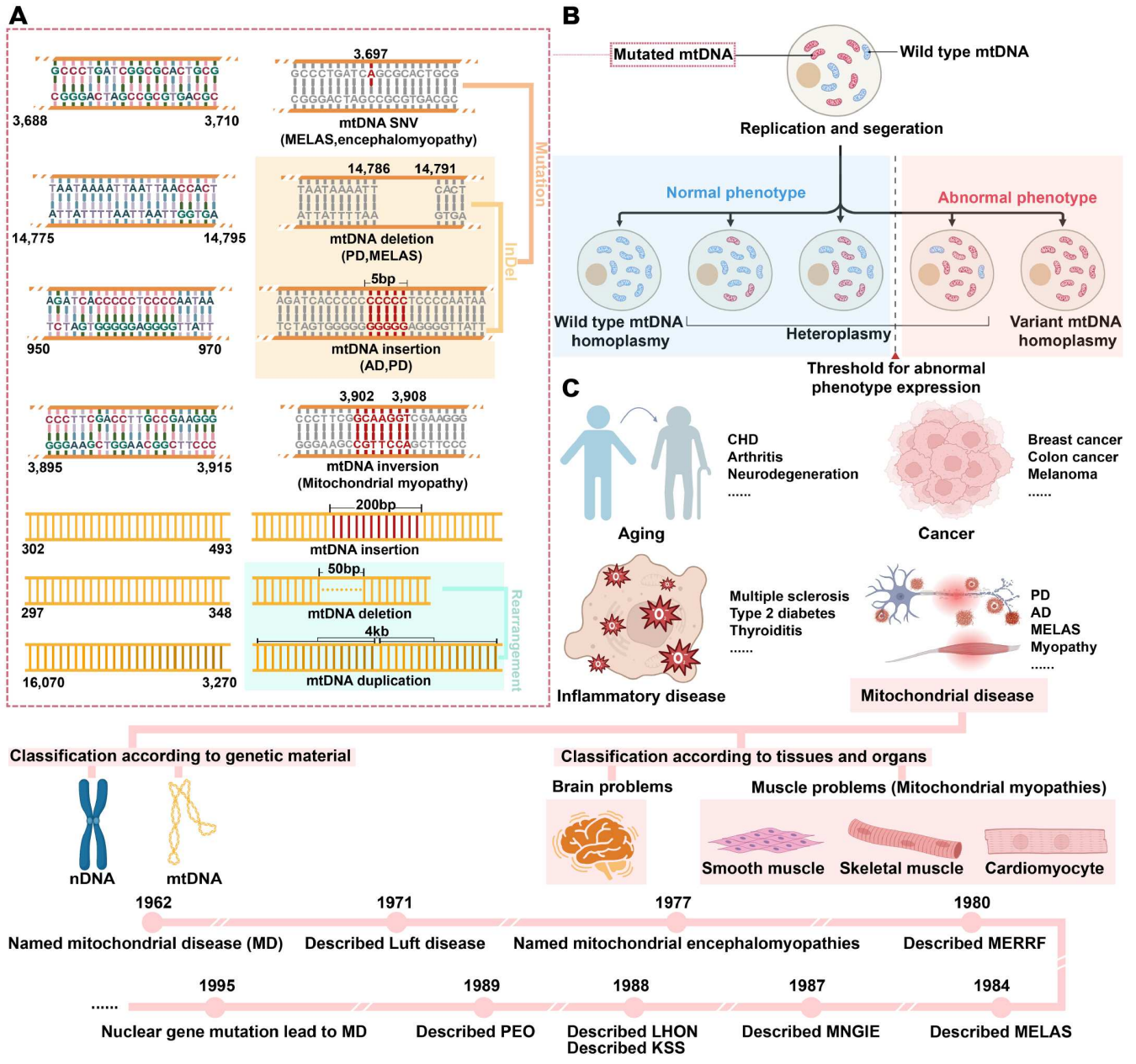


Figure 3. MtDNA variation types and MD categories. A, The red dashed line encloses different types of mtDNA variants. The left column shows the natural mitochondrial sequences referenced to rCRS 17.2, with the position on the mtDNA indicated from 1 to 16569. The right column displays specific positions where variants have been reported, with the base position or length causing each type of variant shown above. MELAS, mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes; PD, Parkinson's disease; AD, Alzheimer's disease. B, The top panel illustrates proliferative human cells containing wild-type or mutated mtDNA. Through replication, cells may produce different ratios of mutated mtDNA to wild-type mtDNA. When the ratio exceeds a certain threshold, the individual may exhibit an abnormal phenotype (Chinnery and Samuels, 1999). C, Main consequences of abnormal phenotypes. Mitochondria play diverse roles in energy metabolism and signal transduction, leading to nonspecific manifestations of abnormal phenotypes. Outcomes may include aging, cancer, inflammatory diseases, and MD (Wallace, 2005). MD represents a spectrum of diseases caused by mitochondrial dysfunction. Although mtDNA variants can lead to MD, they are not solely caused by mtDNA anomalies; abnormalities in nDNA, which are involved in mitochondrial function maintenance, can also result in MD (El-Hattab et al., 2017; Suomalainen and Kaukonen, 2001). MD can be classified by the site of affliction into brain problems, muscle problems, and other issues (Hudson et al., 2007). Conditions presenting exclusively as muscle problems are referred to as mitochondrial myopathies. In most cases, MD involves dysfunction in multiple tissues and is often named after its symptoms. The early research history of MD is illustrated below with a line segment. MERRF: myoclonus epilepsy with ragged-red fibers; MNGIE: myo-, neuro-, gastrointestinal encephalopathy; LHON: Leber hereditary optic neuropathy; KSS: Kearns-Sayre syndrome; PEO: progressive external ophthalmoplegia.

et al., 1999; Ramos et al., 2013; Sondheimer et al., 2011). Animal studies have also demonstrated sex-biased phenomena in heteroplasmy (Fisher and Skibinski, 1990). These findings prompt questions about which type of heteroplasmy is inherited, which is associated with cell proliferation and differentiation, and the extent to which heteroplasmy is influenced by nDNA regulation. The previous study established that severe and pathogenic mtDNA mutations in the germline are predominantly maternally inherited with low frequency, though they can also arise *de novo*. However, the recurrence risk for *de novo* mutations

remains low, emphasizing the critical role of prenatal counseling. The limited cohort size in that study precluded a thorough examination of the differences between maternally inherited and *de novo* mutations (Sallevelt et al., 2017). A recent study has since addressed this gap in understanding. Gupta et al. (2023) analyzed heteroplasmy in approximately 300,000 individuals across various age groups from the UK Biobank and All of Us datasets. After controlling for covariates, they found that heteroplasmic SNVs accumulate with age and tend to arise somatically, whereas heteroplasmic InDels are maternally inherited. Additionally, a genome-wide association study on mtDNA heteroplasmy revealed that multiple nuclear genes regulate heteroplasmy levels and that heteroplasmy across mtDNA sites shares a nuclear genetic architecture (Nandakumar et al., 2021).

MtDNA heteroplasmy is also crucial in the onset of MDs, with pathogenesis differing significantly from that of nDNA variants (Zheng et al., 2011). Although numerous mtDNA mutation sites have been demonstrated to be associated with MDs (Table S1 in Supporting Information), the threshold effect remains a key consideration during diagnosis. This effect refers to the critical level of heteroplasmic mtDNA mutations required to disrupt mitochondrial function and trigger disease symptoms, influencing the onset and severity of MDs (Figure 3B). The threshold is not uniform across individuals and is influenced by factors such as mutation type, location, and the energy demand of different tissues. Consequently, some individuals with high levels of heteroplasmic mutations may be asymptomatic, while others with lower levels exhibit symptoms (Rossignol et al., 2003). Understanding the threshold effect is essential for accurate MD diagnosis, shedding light on the relationship between genetic variants and mitochondrial function. Studies have linked heteroplasmic levels to various diseases, including type 2 diabetes, stroke, and hypertension (Michikawa et al., 1999; Wanagat et al., 2001) (Figure 3C). In many cancers, heteroplasmic variants accumulate (Gopal et al., 2018). The clinical presentations of heteroplasmy can obscure genetic underpinnings, complicating the diagnosis and delaying the treatment of MDs (Elliott et al., 2008). Therefore, understanding heteroplasmic patterns at the population level is imperative. Heteroplasmic levels can be defined relatively (proportion of mutated mtDNA to total mtDNA) or absolutely (quantity of mutated mtDNA), with most research focusing on the former. Previously, it was believed that nonsynonymous heteroplasmic mutations could exist in healthy individuals, but pathogenic heteroplasmy should not. However, this view has changed (Li et al., 2015a; Payne and Chinnery, 2015; Pereira et al., 2011). A pivotal study by Boris et al., analyzing mitogenomes from pedigrees, revealed that on average, each individual carries one type of heteroplasmy, and one in eight individuals harbors a disease-associated heteroplasmy (Rebolledo-Jaramillo et al., 2014). Another study confirmed this conclusion, with 1 in 192 individuals carrying pathogenic heteroplasmic variants without obvious clinical symptoms (Gupta et al., 2023). Single-cell sequencing and large-scale population genomic data focused on mitochondrial genetics revealed dynamic fluctuations in heteroplasmy within individual cells (Glynos et al., 2023; Gupta et al., 2023). These studies suggested that heteroplasmic mutations could be detected in healthy individuals, linking all mitochondrial diseases to heteroplasmy (Gupta et al., 2023). However, all diseases related to heteroplasmy are premature, as recent studies have questioned these correlations (Gupta et al., 2023). After adjusting for unrelated covariates, such as sampling time and blood components, the correlation between heteroplasmic levels and most diseases and sex disappeared. This indicates that the number of heteroplasmic mutations causing abnormal phenotypes is far less than early studies predicted, with many observed differences attributed to changes in cellular composition rather than heteroplasmy levels (Gupta et al., 2023).

This perceptual paradigm shift is largely attributed to the rapid advancement of NGS technology. The earliest Sanger sequencing method had limited applications in complete mitogenome sequencing due to its low throughput, high cost, and challenges in sequencing long repetitive sequences and unique structures. Consequently, early mtDNA sequencing primarily focused on hypervariable regions I (HVR I: 16024 to 16383) and II (HVR II: 57–372), omitting regions with high GC content, homopolymeric tracts, and secondary structural domains. These omissions reduced sequencing resolution, impeding the accurate detection of genetic differences between individuals and the exploration of heteroplasmy within the same individual. Although NGS is prone to false positives or negatives, its enhanced throughput and extended read ranges increase the data density, allowing for more meticulous sequencing of the entire mitogenome and facilitating the detection of minimal levels of heteroplasmy. For instance, a study that sequenced 20 patients with (or suspected of having) mitochondrial cardiomyopathy highlighted the potential of NGS in detecting low-level heteroplasmic mtDNA variants that the Sanger method could not reliably identify (Zaragoza et al., 2010). Another study validated NGS sequencing of mtDNA samples at different mixing ratios, demonstrating a 100% detection rate with virtually no false positives at heteroplasmic levels $\geqslant 5\%$ (Tang and Huang, 2010). Guo et al. (2013a) identified inheritance patterns and bottleneck effects of variants at low heteroplasmic levels in mother-offspring pairs. Subsequent research has employed various quality control methods, gradually lowering the detection limits of heteroplasmy levels to below 1% or even 0.5% (Duan et al., 2019; González et al., 2020). One area worth emphasizing is the study of low-frequency heteroplasmic mutations and aging and reproductive disorders. The relationship between heteroplasmic mutations and aging has been partially elucidated (Figure 3C), yet many complex mechanisms remain to be fully understood (Chomyn and Attardi, 2003; Pinto and Moraes, 2015; Yang et al., 2024). A particularly significant mechanism involves oxidative respiratory byproducts (Lee and Wei, 1997). Mitochondria, during oxidative phosphorylation, produce reactive oxygen species (ROS), which are believed to potentially damage both mtDNA and nDNA (Hartman et al., 2004; Kim et al., 2015; Romesberg and Van Houten, 2022). However, findings from a study employing titratable chemical genetics indicate that ROS released by mitochondria, particularly hydrogen peroxide, is unlikely to cause nDNA damage, thereby minimizing its role in aging and tumorigenesis (van Soest et al., 2024). Similar findings have been corroborated by high-depth sequencing methods. Using an improved Duplex-Seq protocol, significant increases in tissue-specific mtDNA mutations (predominantly G>A/C>T substitutions) have been observed in somatic cells during aging, primarily caused by replication errors and/or cytidine deamination. Notably, G>T/C>A substitutions, which are indicative of oxidative damage, do not show an age-related increase (Sanchez-Contreras et al., 2023). Furthermore, mtDNA mutations have

been linked to reproductive abnormalities, though this remains a subject of debate. In males, variations in mtDNA are often associated with impaired sperm motility and morphology, leading to fertilization failure (Dahadhah et al., 2021; Kao et al., 1995). In females, a conventional view posits that harmful heteroplasmic variations accumulate in germ cells as maternal age increases (Arbeithuber et al., 2022; Rebolledo-Jaramillo et al., 2014; Zaidi et al., 2019). These accumulated mtDNA mutations are thought to impair the NADH/NAD redox function of oocytes, resulting in reproductive abnormalities or disease in offspring (Yang et al., 2020). However, a recent study utilizing DddA-derived cytosine base editors and DNA sequence has established a model showing that increased maternal age enhances the effect of purifying selection, preventing specific heteroplasmic mutations (ND1 3376 G>A and ND5 13513 G>A) from being inherited during oocyte maturation (Ru et al., 2024). These studies are crucial for deepening our understanding of human evolutionary history and genetic diversity, advancing precision medicine and personalized healthcare, and facilitating early disease diagnosis and prevention. While additional empirical validation is necessary, this research provides a fresh perspective for our scientific endeavors.

We have gained a preliminary understanding of the origins, inheritance patterns, genetic regulatory mechanisms, and distribution of heteroplasmy. However, significant differences exist between various types of heteroplasmy, with pathogenic heteroplasmy being widespread in humans (Li et al., 2010). This phenomenon will be explored from an evolutionary perspective on medicine in the following sections. Additionally, some studies suggest that the absolute level of heteroplasmy (mutated mtDNA copy number, mtCN) may more accurately reflect disease severity than the relative level (Filograna et al., 2021). The differences or connections between these factors warrant further investigation.

The patterns and functions of NUMTs

NUMTs are DNA sequences in the nuclear genome that are homologous to mtDNA and originate from the integration of mtDNA fragments into nuclear chromosomes (Stewart et al., 2008). MtDNA-like segments in the nuclear genome were first discovered through hybridization in 1967 (du Buy and Riley, 1967), but it was not until 1994 that Lopez et al. (1994) formally named these fragments NUMTs through sequencing and comparison. A complex and intriguing question arises: Why and how is mtDNA integrated into the nuclear genome? Mitochondria undergo various morphological changes, such as fission, fusion, swelling, shrinking, fragmentation, and autophagy, influenced by genetic regulation or environmental factors. An imbalance in these processes might result in the leakage of mtDNA into the cytoplasm, potentially leading to the transfer of mtDNA fragments into the nuclear genome (Figure 4A and B). Some studies suggest that the interaction between mtDNA and nDNA may be linked to the repair of double-strand breaks (DSBs) in nDNA. The primary mechanisms for DSB repair include homologous recombination and nonhomologous end joining (Baumann and West, 1998; Hazkani-Covo et al., 2010). However, contemporary consensus posits that NUMT formation is exclusively associated with the latter pathway (Figure 4B). This process involves the direct incorporation of free mtDNA bases into DSB termini, facilitated by the coordinated action of proteins such as Ku70/80, DNA-PKcs, Ligase IV, XRCC4, and XLF (Baumann and West, 1998).

Accurate identification of NUMTs is crucial, as misinterpreting them as mtDNA heteroplasmy can lead to erroneous diagnoses of MDs and unreliable interpretations of heteroplasmic variants. Methods to distinguish NUMTs from mtDNA heteroplasmy can be classified into wet laboratory and computational approaches, each with distinct advantages and disadvantages. Wet laboratory methods, such as mitochondrial isolation and PCR primer design for NUMTs, provide intuitive and accurate results (Gould et al., 2015; Ricchetti et al., 2004; Walsh et al., 2022). However, these techniques are expensive and time-consuming and may not capture the full diversity of NUMTs. Consequently, computational methods are more commonly employed. Early approaches utilized the Basic Local Alignment Search Tool (BLAST) to detect and exclude sequences similar to mtDNA from nDNA sequences, primarily focusing on known NUMTs in databases (Parr et al., 2006). Researchers later adopted the use of *K*-mer distribution and frequency to identify potential NUMTs, a method requiring significant computational resources and sensitivity to parameter settings (Li et al., 2019a). Current research often integrates multiple bioinformatics tools to create personalized, efficient, and flexible workflows (Wei et al., 2020). For example, when aligning paired-end sequence data, one end of a read is matched to nDNA and the other to mtDNA; breakpoints are identified by adjusting the reading frame (usually 500 bp) and remapping the identified NUMTs back to nDNA or mtDNA (Ju et al., 2015; Wei et al., 2020). Nevertheless, computational methods are prone to false positives and false negatives. Thus, it is essential to employ statistical measures, such as mtCN, variant allele frequency, and quality scores, to validate the results (Laricchia et al., 2022; Li et al., 2012).

Previous studies have established that NUMTs are prevalent in eukaryotic genomes, with variability in fragment length and number across species (Bensasson, 2001; Uvizl et al., 2024). Some NUMTs originate from entire mitochondrial genomes, indicating complex interactions between mitochondrial and nuclear DNA. Initially, NUMTs were considered remnants of the ancient endosymbiotic process, which occurred approximately two billion years ago. However, Wei et al. (2022) investigated the evolutionary timeline of NUMTs in 66,083 individuals from the UK Biobank and revealed that the emergence of NUMTs has been ongoing throughout human evolution. This finding suggests an essential pathway for intracellular gene transfer and enhances our understanding of the endosymbiotic hypothesis. Furthermore, Wei et al. (2022) refined previous research (Dayama et al., 2014) by mapping the complete landscape of NUMTs. Most NUMTs are nonfunctional and predominantly located in genomic repeat regions, regulatory elements, short interspersed nuclear elements (SINEs), simple sequence repeats, and introns. NUMTs are rarely found in coding regions or within 500 bp of gene transcription start sites (Wei et al., 2022). This distribution may result from the site specificity of DSB repair or the significant selective pressure on coding regions. This study also revealed a negative correlation between the gene intolerance score and the frequency of NUMTs, indicating that NUMTs near genes may be subject to strong evolutionary selection pressures to maintain gene structure and functional stability. A similar study on NUMTs in 45 mammalian species reached comparable conclusions, suggesting that selective pressure on NUMT insertion is a widespread phenomenon (Uvizl

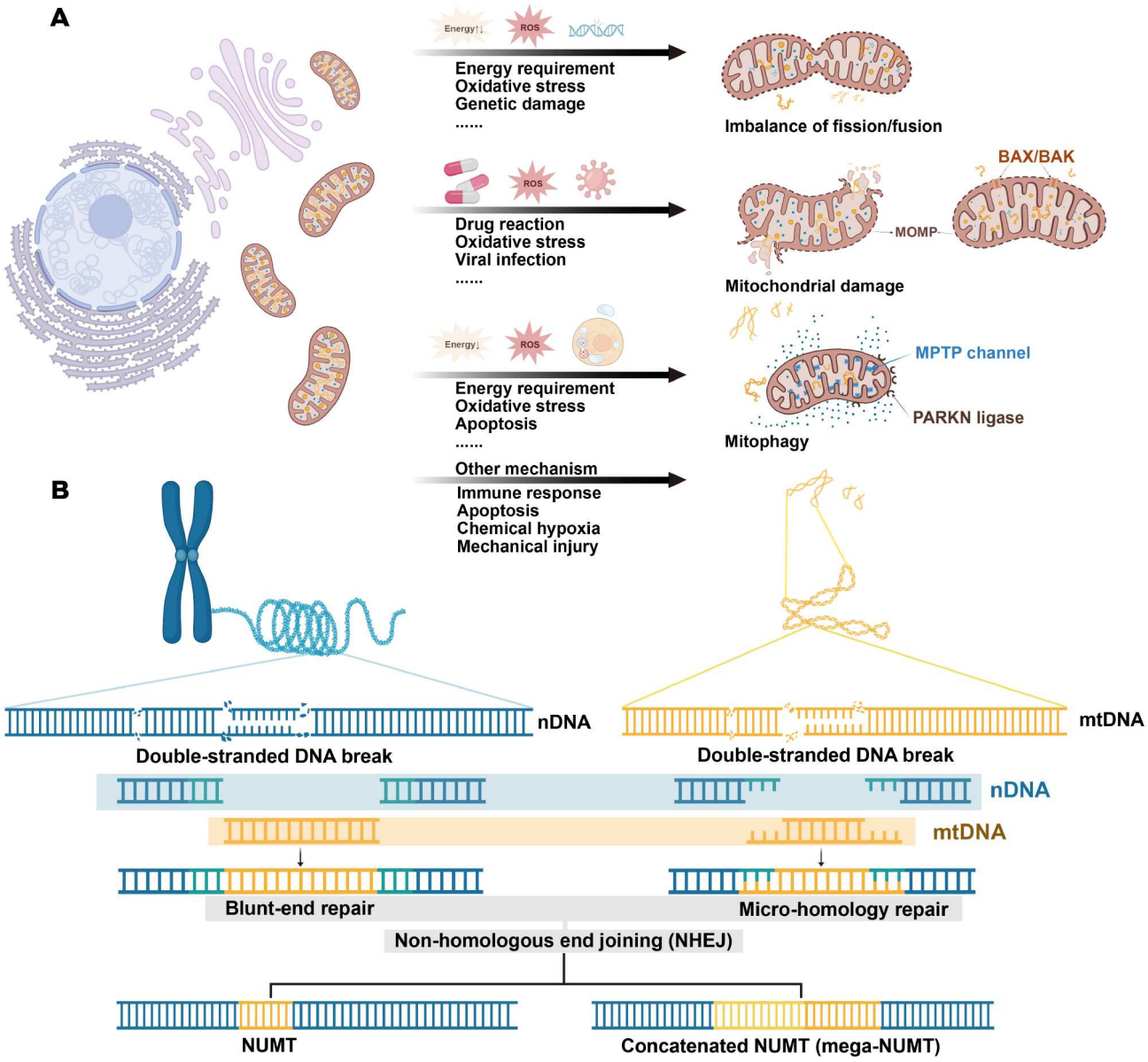


Figure 4. The process of NUMT formation. A, The left panel illustrates a eukaryotic cell with a focus on the nucleus and mitochondria. The cell may encounter various physical, chemical, or biological disruptions (depicted in the middle with text and illustrations), leading to changes in mitochondrial structure. Typical changes are shown on the right, from top to bottom: imbalance of fission/fusion, mitochondrial disruption, and mitophagy (these mechanisms are not entirely independent). B, Structural changes result in the leakage of mtDNA fragments into the cytoplasm (McArthur et al., 2018; Taylor and Turnbull, 2005; West and Shadel, 2017). Simultaneously, the nucleus (chromosomes) may undergo DNA double-strand breaks due to the same factors. During the nDNA repair process, cytoplasmic mtDNA fragments may be used as "sticking-plasters" for repairing damaged nDNA (Wei et al., 2022). Previous studies have identified the main mechanisms involved in nonhomologous end joining (NHEJ), specifically blunt end repair and microhomology repair (Hazkani-Covo et al., 2010). MOMP: mitochondrial outer membrane permeabilization; MPTP: mitochondrial permeability transition pore; PARKN: Parkin RBR E3 ubiquitin protein ligase.

et al., 2024). However, further investigations are needed to determine whether population-specific natural selection occurs within human subgroups and which NUMT fragments are subject to natural selection.

Although NUMTs are primarily nonfunctional, understanding their distribution and dynamic changes across different populations remains significant beyond preventing the misdiagnosis of MDs. The fragment length and quantity of NUMTs vary among populations, potentially reflecting evolutionary histories under diverse genetic backgrounds. NUMTs, which result from complex nuclear-mitochondrial interactions, have critical implications for understanding disease and cancer. Previous studies have suggested that some NUMTs are closely associated with tumorigenesis, potentially indicating that extreme conditions, such as cancer cell proliferation, can trigger NUMT formation or genetic rearrangements (Ju et al., 2015). For example, the *FUS-DDIT3* gene, which is produced by complex rearrangements involving NUMT insertions, is present in 90% of myxoid

liposarcomas, highlighting the carcinogenic role of NUMTs (Göransson et al., 2009). In-depth research on NUMTs is crucial for revealing key pathways of intracellular gene transfer, providing new insights into endosymbiotic theory, and understanding genetic variations and adaptive evolution throughout human history. By analyzing the distribution and dynamics of NUMTs, we can enhance our understanding of the genetic background and evolutionary history of modern humans and their ancestors. This research offers new perspectives and strategies for understanding and precisely diagnosing mitochondria-related diseases and cancers.

The unexpected mode of paternal inheritance

MtDNA is notably conserved across various species, suggesting similarities in biological traits or behaviors. The phenomena of heteroplasmy and NUMTs support this notion. In 1991, Gyllensten et al. (1991) proposed that mtDNA could be paternally inherited based on experiments with inbred mouse strains (C57BL/6J and *Mus spretus*). Similar findings were observed in fruit flies, honeybees, cicadas, chickens, sheep, and cloned cattle (Alexander et al., 2015; Fontaine et al., 2007; Kondo et al., 1992; Meusel and Moritz, 1993; Steinborn et al., 1998; Wolff et al., 2012), prompting further investigation into human mtDNA inheritance patterns. Luo et al. (2018) sequenced three unrelated multigeneration families with high levels of mtDNA heteroplasmy and suggested biparental transmission with an autosomal dominant-like inheritance mode. This study provides evidence for exceptions to paternal inheritance in human mtDNA but could introduce potential complexities in understanding MDs. However, many studies have failed to replicate these findings, leading to skepticism (Carelli and Larsson, 2015; Schwartz and Vissing, 2004; Wei et al., 2022). Pyle et al. (2015) using ~1.2 million-fold coverage of mtDNA sequences in trios, found no evidence of paternal transmission. The debate has gradually reached a consensus with advancements in NUMT research in the NGS era. NUMTs, as part of nDNA, follow an autosomal inheritance pattern, complicating early sequencing results and leading to incorrect conclusions about paternal inheritance (Wei et al., 2020; Wei et al., 2022). Short read lengths can cause assembly difficulties, reduce mapping accuracy, and hinder the identification of NUMTs. Therefore, an optimized pipeline for NUMT detection and high-coverage WGS data, especially from PacBio SMRT or Oxford Nanopore technologies, is essential for distinguishing NUMTs from true mtDNA variants. The debate over paternal inheritance of human mtDNA underscores the importance and potential of NGS and TGS technologies in genetic research. These technologies can reduce systematic errors, thereby deepening our understanding of the distribution, evolution, and function of NUMTs in the genome. Although the specific roles of NUMTs in species evolution and disease progression remain unclear, innovations in sequencing technologies and computational biology algorithms hold promise for providing novel insights into these questions.

Maternal evolutionary history and genetic legacy of mtDNA

Evolutionary history is essential for understanding the origins, migrations, admixture, and genetic diversity patterns of ethnolinguistically diverse human populations. This knowledge also aids in comprehending the traits and disease susceptibility of genetically distinct groups. MtDNA, as an ancient and maternally inherited marker, has accumulated sufficient mutations over time to be used to distinguish genetically different populations. The position and number of these mutations can determine the differences between groups, thereby elucidating the phylogenetic relationships among populations. This discussion reviews the significant contributions made by technological advancements in this field.

In 1987, Cann et al. (1987) sequenced and analyzed 370 mtDNA restriction sites in 147 individuals globally, revealing that the most recent common maternal ancestor of modern humans outside of Africa, known as "mitochondrial Eve", originated from a relatively small group of the African population. This study provided a broad overview of the genetic makeup of modern humans across different continents but lacked a detailed understanding of the genetic structure of geographically distinct populations (Cann et al., 1987). Although HVRs are mutation-rich, they constitute only a small portion of the mitogenome, rendering partial sequencing insufficient for reconstructing a detailed phylogeny and exploring the fine-scale genetic structure among populations. NGS has addressed these challenges, enabling discoveries and a deeper understanding of genetic diversity. For example, early targeted sequencing data suggested that a single out-of-Africa event populated Australasia and Eurasia, given the similar coalescence times of the three macro-haplogroups (M, N, and R) in these regions (Forster, 2004; Macaulay et al., 2005). However, whole-genome sequencing revealed significant differences in the divergence times between aboriginal Australians and Eurasian populations approximately 70,000 years ago and 30,000 years ago, respectively, refuting the earlier hypothesis (Rasmussen et al., 2011). Whole-mtDNA sequencing has also highlighted the rapid migration of modern humans from Africa along the "southern route" to Asia (Kong et al., 2006). Subsequent studies refined this route to include expansions from southern China to the Himalayas and from Southeast Asia along river systems into East Asia (Li et al., 2015b; Peng et al., 2011). Additionally, whole-genome sequencing of 19,568 mtDNA sequences from northern Pakistani and surrounding populations has elucidated the genetic impact of Yamnaya-related steppe populations on South Asian populations (Rahman et al., 2021) and the deep connections between matrilineal ancestry from northern East Asia, Japan, and the Americas (Li et al., 2023). Multiple studies have detailed human migration histories and clarified the genetic diversity that was formed through complex population admixture processes. For instance, whole mtDNA sequences of Tibetan populations confirmed their composition of Paleolithic maternal components and Neolithic millet farmers from the Yellow River Basin, refining previous archaeological views (Li et al., 2019b; Zhao et al., 2009). Similarly, indigenous populations in southern China and Southeast Asia exhibited ancient genetic components as well as genetic components that have formed since the Holocene (Kong et al., 2011). Generally, whole-genome sequencing is particularly advantageous for analyzing sex-biased admixture. A comparative study of patrilocal and matrilocal groups in Sumatra revealed that matrilocality leads to lower maternal genetic diversity within lineages. This difference was not detected by earlier HVR sequencing (Gunnarsdóttir et al., 2011). Advances in sequencing technology also offer significant advantages in resolving issues of demographic and cultural

diffusion, especially in the complex genetic and cultural landscape of East Asians. For example, geographically diverse Tai-Kadai groups exhibit high genetic homogeneity and cultural similarity across East and Southeast Asia, aligning with the demic diffusion model (Kutanan et al., 2017). In contrast, the Han Chinese formation is more complex, with a clear north–south divide formed by the cultural conquest and assimilation of southern indigenous populations, supporting the cultural diffusion model (Yao et al., 2002). In summary, the critical role of whole mtDNA sequencing based on NGS in resolving the issues of population origin, divergence, migration, and admixture is widely recognized. Recent advancements in ancient DNA extraction and sequencing techniques have further corroborated or revised many hypotheses, enhancing our understanding of population divergence and genetic connections (Nesheva, 2014; Haber et al., 2016). A compelling example is provided by ancient DNA evidence from Germany, Sweden, and Luxembourg, which indicates that the genetic component of Anatolian farmers in modern European populations is derived from Sardinia in western Italy rather than from Southeast Europe, as suggested by the demic diffusion model (Keller et al., 2012; Lazaridis et al., 2014; Skoglund et al., 2012). Similarly, the origin of Austronesians in Asia has been a topic of intense scholarly debate. Some experts propose the Taiwan island of coastal China as the cradle of this group, while others suggest inland South China. Ancient DNA findings have effectively remapped the early migratory path of Austronesians from South China through Taiwan, leading to their expansive distribution across the Southeast Asian archipelago, Madagascar, and the broader Oceanic realm (Ko et al., 2014). Additionally, ancient DNA has illuminated the genetic components of archaic humans and their interactions with early modern humans (Vernot et al., 2021; Zhang et al., 2020). These findings not only enhance our understanding of early human history and migration patterns but also provide new insights into the genetic differences among modern human populations.

From an evolutionary perspective, human diseases and phenotypic variations result from constraints, trade-offs, mismatches, and conflicts inherent in complex biological systems interacting with changing environments through natural selection (Benton et al., 2021). This study aimed to elucidate this viewpoint by examining mitochondrial mutations. An intriguing question arises when considering that mutations define different haplogroups: do these mutations lead to functional differences or diseases that affect organisms' adaptation or selection in specific environments, thereby influencing the current distribution of populations? Some studies have attempted to answer this question. Research on nonsynonymous and synonymous mutations in the mtDNA coding region has shown that ATP6, cytochrome b, and cytochrome c oxidase I (genes that play significant roles in OXPHOS) exhibit considerable functional differences across populations at different latitudes (Mishmar et al., 2002), suggesting a potential link between mutations and the distribution of genetically diverse populations (Ruiz-Pesini et al., 2004). Specifically, the haplogroups M and N predominate in Asia, with the northern haplogroup N characterized by two amino acid variants (ND3 and ATP6). These variants alter mitochondrial membrane potential and Ca^{2+} levels (Kazuno et al., 2006), affecting coupling efficiency and increasing thermogenesis, which may offer an advantage in the adaptation of cold environments. Studies have shown that haplogroups A, C, and D, which are prevalent in Siberia, have a higher basal metabolic rate than southern populations belonging to haplogroups B and F (Leonard et al., 2002; Snodgrass et al., 2008; Snodgrass et al., 2005). MDs also display unique distribution patterns (Table S2 in Supporting Information). For instance, Leber's hereditary optic neuropathy (LHON), linked to mutations at positions 11778 or 14484, has a greater disease risk in the widely distributed European haplogroup J than in other haplogroups (Brown et al., 1997; Brown et al., 1995; Johns et al., 1992; Torroni et al., 1997). Additionally, in the genetic background of haplogroups U and J, mutations at positions 16126 and 73 are more likely to cause age-related macular degeneration and retinal pigment issues (Jones, 2007; Udar et al., 2009). Furthermore, Russian Tatar populations belonging to haplogroup H may increase the risk of Parkinson's disease (Khusnutdinova et al., 2008), whereas populations with haplogroups U and K show a protective effect (Ghezzi et al., 2005; van der Walt et al., 2004; van der Walt et al., 2003). The mechanisms and environmentally dependent impacts linking mutations to phenotypic traits or disease susceptibility are complex and variable. For example, the 3394 T>C mutation associated with LHON decreases complex I activity in haplogroups B and F, reducing oxidative respiration. However, this mutation frequently occurs in populations on the Tibetan Plateau (haplogroup M9) (Ji et al., 2012). The pleiotropy of mitochondrial genes further complicates these associations. Despite the higher risk of myocardial infarction in East Asian populations with haplogroup N9a, it is simultaneously associated with a lower risk of diabetes (Fuku et al., 2007; Nishigaki et al., 2007). In the Han Chinese population, haplogroup M7b1a1 is linked to a lower BMI, waist circumference, and waist-to-hip ratio, indicating a reduced risk of obesity. However, it also leads to higher levels of total cholesterol and low-density lipoprotein cholesterol (Chen et al., 2024).

In summary, preliminary findings on mitochondrial mutations and their impacts on population adaptability and disease susceptibility in specific environments illustrate the role of genetic variations within complex environmental and biological networks. To delve deeper into the intricate relationships among genetics, phenotypes, and population dynamics throughout evolutionary processes, there has been a shift from traditional population-based research models to more advanced biobank-based approaches. This transition underscores the significance of advanced computational tools and bioinformatics algorithms. These tools enable detailed exploration of evolutionary dynamics such as population differentiation, gene flow, and genetic drift, thereby addressing various biological, medical, and archaeological issues (Balloux et al., 2009; Pääbo et al., 2004). Furthermore, statistical modeling is crucial for integrating genomic data, quantifying genetic diversity, testing hypotheses about population differentiation and admixture processes, and understanding the genetic basis of adaptive evolution. For example, Bayesian statistical methods and maximum likelihood estimation based on mtDNA sequence variation facilitate the analysis of population differentiation timing and rate, revealing patterns of historical migration and expansion (Beaumont, 1999; Fagundes et al., 2007). Overall, integrating ancient and modern genomic data enables the exploration of the historical spread of specific mtDNA variants and their effects on the genetic makeup of contemporary populations. This approach also helps determine how genetic diversity reflects ancient population divergence and migration events and how mtDNA-based genetic adaptability evolves under specific environmental pressures (Excoffier and Lischer, 2010).

Sequencing progression, genomic databases, and analytical techniques

Accurate identification of individual heteroplasm levels, elucidation of the formation mechanisms of NUMTs, reshaping of evolutionary history, and advancements in personalized medicine and precision treatments depend on the collection, storage, and analysis of biological data. This presents challenges in sequencing technologies, databases, and computational methods (He et al., 2024; Wang et al., 2024a; Wang et al., 2024b). The paradigm shift brought by NGS has significantly enhanced our understanding (Metzker, 2009), yet this approach has notable limitations. Distinguishing true heteroplasmic variants from false positives caused by NUMTs remains challenging. PCR amplification of DNA samples before NGS can introduce bias, affecting the sensitivity of low-level heteroplasmy detection and increasing the detection limit. Moreover, the short reads produced by NGS do not facilitate the sequencing of the entire mitogenome in one step, nor do they effectively identify long NUMT segments or variants in complex genomic regions such as repetitive sequences. This complicates accurate mapping, especially in the presence of numerous NUMTs, leading to potential errors. In contrast, TGS offers longer reads without PCR amplification, improving the accuracy of detecting low-level heteroplasmy, structural variations, and NUMTs despite several limitations, such as higher error rates and throughput constraints. To fully leverage TGS technology, two pressing needs must be addressed: establishing high-quality genomic biobank databases encompassing diverse ethnolinguistic groups and comprehensive disease spectra and developing optimized algorithms for swiftly and accurately identifying homoplasmic mtDNA variants, NUMTs, and heteroplasmy.

Various studies have underscored the significance and challenges of establishing a comprehensive mtDNA biobank for human genetics, evolution, and disease diagnosis. These challenges include sequencing technologies, dataset scale, database quality, and accessibility. The most critical limitation of sequencing technology is that it often compromises mtDNA data quality through five types of errors: base shifts, reference bias, phantom mutations, base mismatching, and artificial recombination (Bandelt et al., 2004). These errors can lead to incorrect sequence alignments and interpretations, particularly in complex regions. Early solutions that ignored complex variants resolved some alignment errors but overlooked significant sequence and length polymorphisms, thus failing to decipher fine-scale population genetic structures. In addition to sequencing technology, the demands of various application fields have hindered the development of a representative population biobank database (Figure 5E and F) (Kogelnik, 1996; Parson and Dür, 2007; Preste et al., 2018; Ratnaike et al., 2021; Rubino et al., 2011; Wittig et al., 2003). For instance, databases such as MitoPhen and HmtDB focus primarily on MDs, leading to inadequate data quality control for nonpathogenic variants. Evolutionary scientists, on the other hand, require accurate and comprehensive data on mutation rates and allele or haplogroup frequencies to characterize mtDNA genetic diversity in global populations. Databases containing only HVR information, such as D-loop-base and HmtDB, are insufficient for these needs. European bias in large-scale genomic projects is another critical issue that is prevalent in multiple biobanks and genetic databases (Li et al., 2024; Sirugo et al., 2019; Sun et al., 2024; Wang et al., 2024b). For example, the EMPOP (European DNA Profiling Forensic mtDNA Population Database) and MITOMAP (Mitochondrial MAP) databases predominantly store mitogenome data from European populations. Even the widely used reference sequence (rCRS) is of European origin (Figure 5F). This bias neglects significant genetic variants in specific populations, affecting disease diagnosis, treatment, and prevention strategies, and may result in incorrect interpretations of ancient population migrations, evolution, and forensic identification, thus limiting the general applicability of these databases. Aside from population bias, accessibility and maintainability are crucial for evaluating the usefulness of databases. D-loop-BASE, for example, is not accessible via the internet, and individual sequences are not published, raising questions about data quality (Bandelt and Parson, 2004). Although Lincoln and Carracedo argue that data should be provided based on the authors' requirements, this is rarely achieved (Lincoln and Carracedo, 2000). Based on these observations, we propose that an ideal mtDNA database should combine characteristics such as comprehensiveness, high quality, broad applicability, representativeness, accessibility, and transparency to support the broad needs of evolutionary genetics, medical research, and forensic investigations.

In the genomic era, an increasingly challenging issue is the processing and analysis of the immense volume of data generated (Schadt et al., 2010). Analyzing such vast amounts of data requires significant computational resources and complex bioinformatics tools. A critical challenge is aligning sequences of variable lengths, and numerous tools and pipelines have been developed to address this challenge (Figure 5C and D). NGS has shifted the research focus from merely identifying presence or absence to quantifying variations, imposing new requirements for software to detect heteroplasmic levels. Generally, all software programs use base quality assessment, read alignment, and heteroplasmic level calculations. We summarized some popular tools currently available, noting their limitations (Calabrese et al., 2014; Guo et al., 2013b; Vellarikkal et al., 2015; Weissensteiner et al., 2016; Zhidkov et al., 2011) (Figure 5A and B). For instance, MitoBamAnnotator accepts only single-end sequence input, potentially missing heteroplasmy data from other DNA strands. Despite the versatility of the mit-o-matic with web and CLI interfaces in catering to diverse analytical needs, it is hampered by an excess of parameter options, often leading to suboptimal and unreliable outcomes. Furthermore, integrated pipelines using various software through the CLI demand manual configuration of a comprehensive runtime environment pose significant challenges and reduce the efficiency of data transfer, analysis, and interpretation.

Significant progress has been made in constructing high-quality biobanks and developing corresponding software, yet this field still faces numerous challenges. These include interpreting and managing signal noise or information loss due to technical limitations in large-scale genomic data and linking genetic information with complex real-world environments. Artificial intelligence (AI) offers promising solutions to these challenges. Machine learning, a branch of AI, enables computers to enhance the efficiency and accuracy of data processing by learning from data and algorithms without explicit programming (Vieira et al., 2020). This includes techniques for automatically identifying patterns in data, which can be used for predictions or decisions (Russell and Norvig, 2010). Deep learning technologies have demonstrated great potential in identifying patterns and signals

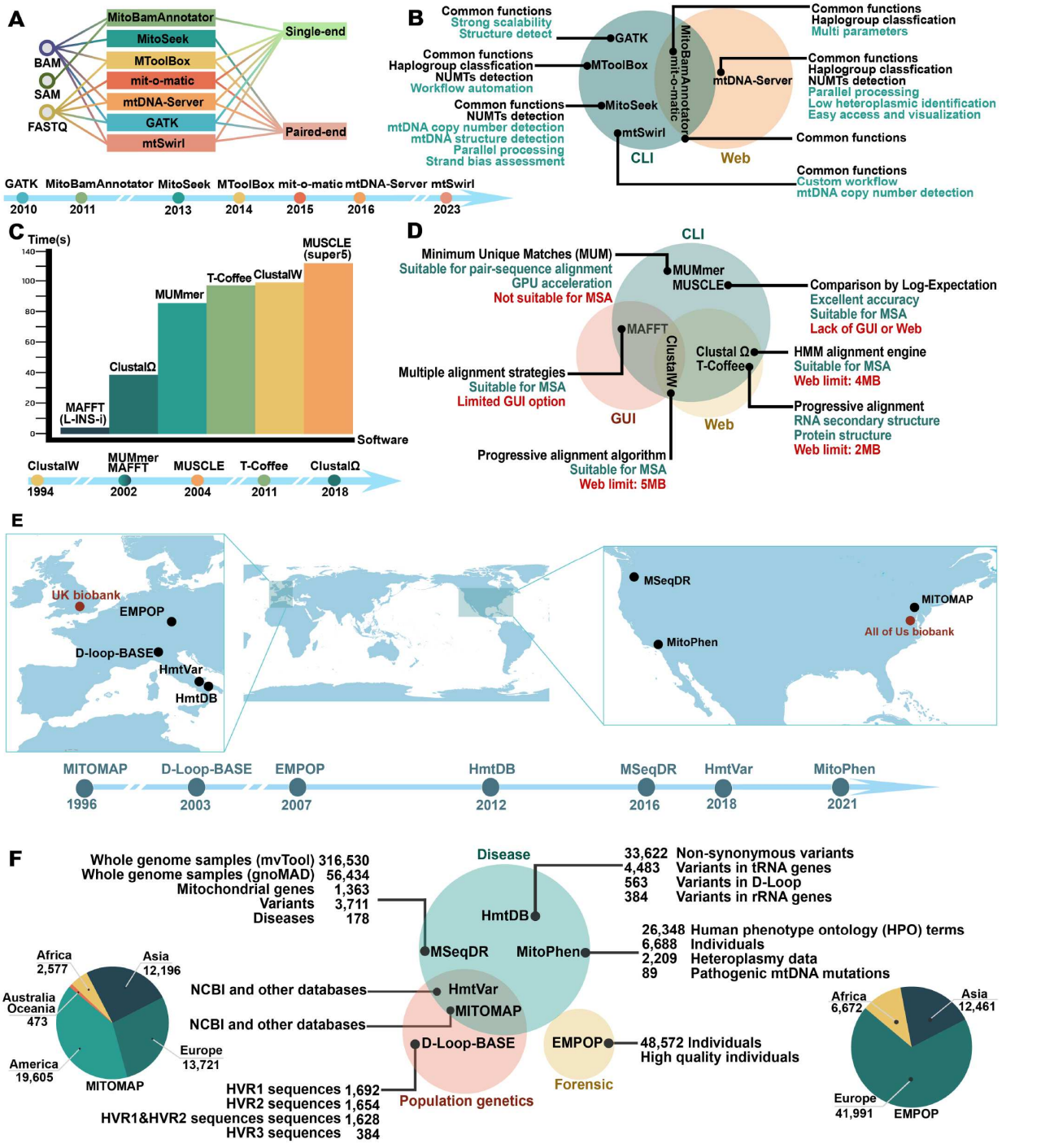


Figure 5. Features of mitogenome alignment tools and mtDNA datasets. A, Main tools for mtDNA analysis and the data types they can process (Calabrese et al., 2014; Guo et al., 2013b; Vellarikkal et al., 2015; Weissensteiner et al., 2016; Zhidkov et al., 2011). Tools (rectangles) are connected to the file formats they handle (circles) with lines. The timeline below depicts the release years of these tools. B, Comparative chart of the main features of the tools mentioned in a. Tools are divided into command-line interfaces (CLIs) and web-based interfaces (WWs). All tools perform heteroplasmy detection and variant annotation. Unique features are indicated in green font. C, Histogram showing the speed of multiple sequence alignment by mainstream alignment software (Di Tommaso et al., 2011; Edgar, 2004; Katoh, 2002; Kurtz et al., 2004; Sievers and Higgins, 2017; Thompson et al., 1994). Tests were conducted on an i7-13700K, 16 GB RAM Linux system with ten different human mtDNA sequences using the fastest alignment algorithm available for each software (indicated in parentheses). The release years of the software are shown on a timeline below. D, Characteristics of the alignment software from c. Tools are divided into CLI, Web, and graphical user interface (GUI) based on the operating platform. The algorithm used by each software is indicated in black font. The advantages are indicated in green font, and the limitations are indicated in red. E, Map showing the distribution of the mtDNA databases (Kogelnik, 1996; Parson and Dür, 2007; Preste et al., 2018; Ratnaike et al., 2021; Rubino et al., 2011; Shen et al., 2016; Wittig et al., 2003). The left and right maps are magnified views of Europe and the USA, respectively. The UK Biobank and All of Us databases are included due to their large amount of mtDNA data content. The establishment years of the databases are shown on a timeline below, connected to the map with dashed lines. F, Comparison of mtDNA database inclusion, as shown in e. Databases are categorized by purpose: disease, population genetics, and forensics. Pie charts displaying the number of individuals included in the EMBASE and MITOMAP databases. HmtVar and HmtDB have not been updated since 2018, and D-loop-BASE is not publicly accessible. Overall, the quality of current mtDNA databases is not satisfactory.

from complex data (LeCun et al., 2015; Li et al., 2022). For instance, Min et al. (2016) used deep learning methods to analyze human genome data and successfully predicted the genetic risk of various complex diseases. Angermueller et al. (2016) accurately identified expression patterns related to specific diseases through deep learning models analyzing gene expression data. Poplin et al. (2018) developed a high-accuracy variant detection tool using deep learning methods to identify SNVs and InDels from high-throughput sequencing data, providing an effective means for genetic research and variant interpretation. Additionally, ensemble learning methods can improve prediction accuracy and robustness by combining multiple models (Dietterich, 2000). Enhancing the interpretability of algorithms is another critical direction in AI and machine learning development. This enables algorithms to provide insights into their decision-making processes, aiding in understanding prediction outcomes and making informed decisions (Holzinger et al., 2019). The development of models capable of processing heterogeneous data and considering the interactions between genetic information and the real world is vital for solving problems in complex biological systems (Libbrecht and Noble, 2015). Machine learning and AI are expected to play increasingly important roles in deciphering genetic information, advancing personalized medicine, and understanding the genetic basis of human diseases. Interdisciplinary collaboration involving computational science and genetics is essential to achieve these goals (Topol, 2019; Zou et al., 2018).

Conclusions and perspectives

This review revisits recent advancements in mtDNA research within medical science and evolutionary genomics facilitated by NGS technologies. We emphasize the impact of heteroplasmy on mitochondrial function and its association with MDs, as well as the role of NUMTs in challenging traditional perceptions of mtDNA inheritance and enhancing our understanding of the endosymbiotic hypothesis. Additionally, we highlight how large-scale population mtDNA studies and ancient DNA sequencing techniques reveal humanity's extensive migration paths and adaptations to diverse environments, shaping our current maternal genetic structure. Moreover, this paper examines the characteristics of contemporary computational tools that improve multiple sequence alignments, enhancing variant identification accuracy and deepening our understanding of the relationships between structure and function. These advancements refine our knowledge of the roles played by mtDNA variants in biological processes. Finally, we underscore the importance of establishing a comprehensive, high-quality, and accessible mtDNA database encompassing both common and rare genetic variants.

Leveraging TGS technologies, characterized by long reads, single-molecule sequencing, and high-resolution structural variation detection, is pivotal for achieving our research objectives. This approach facilitates the detection of minimal heteroplasmy levels and enhances the identification of NUMT distributions, enriching our understanding of MDs and mtDNA inheritance patterns. Additionally, the ability of TGS to identify chemical modifications of mtDNA may reveal epigenetic alterations, providing insights into the effects of TGS on mitochondrial functionality and cellular metabolism. This advancement broadens our understanding of the interplay between MDs, genetics, and environmental factors.

The widespread application of TGS is essential for the Human Pangenome Project's mission to map all human genetic variations, including the core genome shared by diverse populations, the accessory genome found in select individuals, and the unique genome specific to each person (Wang et al., 2022). This project transcends the limitations of single-reference genome research, enabling a more comprehensive capture of genetic variation, particularly rare variations, structural variants, and their phenotypic implications. By comparing accessory and unique genomes, we can trace the emergence and dissemination of variations, elucidating how individuals adapt to environmental changes through gene addition, loss, or modification. This process reveals the intricate genetic fabric within populations and investigates genetic differences and evolutionary paths across groups, offering invaluable resources for disease diagnosis and personalized medicine amidst the complexity of human genetics. Pangenome-based analyses across various populations and species may uncover significant mitochondrial-nuclear linkages, clarifying the timing and reasons for their correlation and facilitating a deeper exploration of endosymbiosis and the dynamics and mechanisms underlying NUMTs. In conclusion, we anticipate that future breakthroughs in human genetic research will revolutionize medical research, disease management, and anthropological inquiries. Achieving this goal requires interdisciplinary collaboration across evolutionary genomics, bioinformatics, computer science, and other fields to overcome challenges related to sample bias, data processing, and analysis.

Compliance and ethics

The authors declare that they have no conflict of interest. Figures 3 and 4 (Agreement Nos: GH2738EN6F and AV2738EWG9) were created using BioRender.com (https://www.biorender.com/).

Acknowledgement

This study was supported by the National Natural Science Foundation of China (82202078 and 82402203), the National Natural Science Foundation of Chongqing (CSTB2024NSCQ-LZX0005), the Major Project of the National Social Science Foundation of China (23&ZD203), the Open Project of the Key Laboratory of Forensic Genetics of the Ministry of Public Security (2022FGKFKT05), the Center for Archaeological Science of Sichuan University (23SASA01), the 1.3.5 Project for Disciplines of Excellence, West China Hospital, Sichuan University (ZYJC20002), and the Sichuan Science and Technology Program (2024NSFSC1518).

Supporting information

The supporting information is available online at https://doi.org/10.1007/s11427-024-2736-7. The supporting materials are published as submitted, without typesetting or editing. The responsibility for scientific accuracy and content remains entirely with the authors.

References

Alexander, M., Ho, S.Y.W., Molak, M., Barnett, R., Carlborg, Ö., Dorshorst, B., Honaker, C., Besnier, F., Wahlberg, P., Dobney, K., et al. (2015). Mitogenomic analysis of a 50-generation chicken pedigree reveals a rapid rate of mitochondrial evolution and evidence for paternal mtDNA inheritance. Biol Lett 11, 20150561.

Anderson, S., Bankier, A.T., Barrell, B.G., de Bruijn, M.H.L., Coulson, A.R., Drouin, J., Eperon, I.C., Nierlich, D.P., Roe, B.A., Sanger, F., et al. (1981). Sequence and organization of the human mitochondrial genome. Nature 290, 457–465.

Angermueller, C., Pärnamaa, T., Parts, L., and Stegle, O. (2016). Deep learning for computational biology. Mol Syst Biol 12, 878.

Arbeithuber, B., Cremona, M.A., Hester, J., Barrett, A., Higgins, B., Anthony, K., Chiaromonte, F., Diaz, F.J., and Makova, K.D. (2022). Advanced age increases frequencies of *de novo* mitochondrial mutations in macaque oocytes and somatic tissues. Proc Natl Acad Sci USA 119, e2118740119.

Balloux, F., Handley, L.J.L., Jombart, T., Liu, H., and Manica, A. (2009). Climate shaped the worldwide distribution of human mitochondrial DNA sequence

variation. Proc R Soc B 276, 3447–3455.
Bandelt, H.J., Salas, A., and Bravi, C. (2004). Problems in FBI mtDNA database. Science 305, 1402–1404.
Bandelt, H.J. (2005). More evidence for non-maternal inheritance of mitochondrial DNA? J Med Genet 42, 957–960.
Bandelt, H.J., and Parson, W. (2004). Fehlerquellen mitochondrialer DNS-Datensätze und Evaluation der mtDNS-Datenbank? D-Loop-BASE? Rechtsmedizin 14, 251–257.
Baumann, P., and West, S.C. (1998). Role of the human RAD51 protein in homologous recombination and double-stranded-break repair. Trends Biochem Sci 23, 247–251.
Beaumont, M.A. (1999). Detecting population expansion and decline using microsatellites. Genetics 153, 2013–2029.
Bensasson, D. (2001). Mitochondrial pseudogenes: evolution's misplaced witnesses. Trends Ecol Evol 16, 314–321.
Benton, M.L., Abraham, A., LaBella, A.L., Abbot, P., Rokas, A., and Capra, J.A. (2021). The influence of evolutionary history on human health and disease. Nat Rev Genet 22, 269–283.
Brown, M.D., Sun, F., and Wallace, D.C. (1997). Clustering of Caucasian Leber hereditary optic neuropathy patients containing the 11778 or 14484 mutations on an mtDNA lineage. Am J Hum Genet 60, 381–387.
Brown, M.D., Torroni, A., Reckord, C.L., and Wallace, D.C. (1995). Phylogenetic analysis of Leber's hereditary optic neuropathy mitochondrial DNA's indicates multiple independent occurrences of the common mutations. Hum Mutat 6, 311–325.
Calabrese, C., Simone, D., Diroma, M.A., Santorsola, M., Guttà, C., Gasparre, G., Picardi, E., Pesole, G., and Attimonelli, M. (2014). MToolBox: a highly automated pipeline for heteroplasmy annotation and prioritization analysis of human mitochondrial variants in high-throughput sequencing. Bioinformatics 30, 3115–3117.
Cann, R.L., Stoneking, M., and Wilson, A.C. (1987). Mitochondrial DNA and human evolution. Nature 325, 31–36.
Carelli, V., and Larsson, N.G. (2015). Keeping in shape the dogma of mitochondrial DNA maternal inheritance. PLoS Genet 11, e1005179.
Carelli, V., Ross-Cisneros, F.N., and Sadun, A.A. (2004). Mitochondrial dysfunction as a cause of optic neuropathies. Prog Retinal Eye Res 23, 53–89.
Chen, Z., Chen, L., Tan, J., Mao, Y., Hao, M., Li, Y., Wang, Y., Li, J., Wang, J., Jin, L., et al. (2024). Natural selection shaped the protective effect of the mtDNA lineage against obesity in Han Chinese populations. J Genet Genomics S1673-8527(24) 00129-2.
Chinnery, P.F., and Samuels, D.C. (1999). Relaxed replication of mtDNA: a model with implications for the expression of disease. Am J Hum Genet 64, 1158–1165.
Chomyn, A., and Attardi, G. (2003). MtDNA mutations in aging and apoptosis. Biochem Biophys Res Commun 304, 519–529.
Comas, D., Pääbo, S., and Bertranpetit, J. (1995). Heteroplasmy in the control region of human mitochondrial DNA.. Genome Res 5, 89–90.
Nesheva, D. (2014). Aspects of ancient mitochondrial DNA analysis in different populations for understanding human evolution. Balkan J Med Genet 17, 5–14.
Dahadhah, F.W., Jaweesh, M.S., Al Zoubi, M.S., Alarjah, M.I.A., Hammadeh, M.E., and Amor, H. (2021). Mitochondrial nicotinamide adenine dinucleotide hydride dehydrogenase (NADH) subunit 4 (MTND4) polymorphisms and their association with male infertility. J Assist Reprod Genet 38, 2021–2029.
Dayama, G, Emery, S. B., Kidd, J. M, and Mills, R. E (2014). The genomic landscape of polymorphic human nuclear mitochondrial insertions. Nucleic Acids Res 42, 12640–12649.
Di Tommaso, P., Moretti, S., Xenarios, I., Orobitg, M., Montanyola, A., Chang, J.M., Taly, J.F., and Notredame, C. (2011). T-Coffee: a web server for the multiple sequence alignment of protein and RNA sequences using structural information and homology extension. Nucleic Acids Res 39, W13–W17.
Dietterich, T.G. (2000). Ensemble methods in machine learning. Multiple Classifier Systems. Berlin, Heidelberg: Springer.
du Buy, H.G., and Riley, F.L. (1967). Hybridization between the nuclear and kinetoplast DNA's of *Leishmania enriettii* and between nuclear and mitochondrial DNA's of mouse liver. Proc Natl Acad Sci USA 57, 790–797.
Duan, M., Chen, L., Ge, Q., Lu, N., Li, J., Pan, X., Qiao, Y., Tu, J., and Lu, Z. (2019). Evaluating heteroplasmic variations of the mitochondrial genome from whole genome sequencing data. Gene 699, 145–154.
Edgar, R.C. (2004). MUSCLE: a multiple sequence alignment method with reduced time and space complexity. BMC BioInf 5, 113.
Ekstrand, M.I. (2004). Mitochondrial transcription factor A regulates mtDNA copy number in mammals. Hum Mol Genet 13, 935–944.
El-Hattab, A.W., Craigen, W.J., and Scaglia, F. (2017). Mitochondrial DNA maintenance defects. Biochim Biophys Acta Mol Basis Dis 1863, 1539–1555.
Elliott, H.R., Samuels, D.C., Eden, J.A., Relton, C.L., and Chinnery, P.F. (2008). Pathogenic mitochondrial DNA mutations are common in the general population. Am J Hum Genet 83, 254–260.
Excoffier, L., and Lischer, H.E.L. (2010). Arlequin suite ver 3.5: a new series of programs to perform population genetics analyses under Linux and Windows. Mol Ecol Resour 10, 564–567.
Fagundes, N.J.R., Ray, N., Beaumont, M., Neuenschwander, S., Salzano, F.M., Bonatto, S.L., and Excoffier, L. (2007). Statistical evaluation of alternative models of human evolution. Proc Natl Acad Sci USA 104, 17614–17619.
Filograna, R., Mennuni, M., Alsina, D., and Larsson, N.G. (2021). Mitochondrial DNA copy number in human disease: the more the better? FEBS Lett 595, 976–1002.
Filosto, M., Mancuso, M., Vives-Bauza, C., Vilà, M.R., Shanske, S., Hirano, M., Andreu, A.L., and DiMauro, S. (2003). Lack of paternal inheritance of muscle mitochondrial DNA in sporadic mitochondrial myopathies. Ann Neurol 54, 524–526.
Fisher, C., Skibinski, D. O. F. (1990). Sex-biased mitochondrial DNA heteroplasmy in the marine mussel *Mytilus*. Proc Roy Soc London Ser Biol Sci 242, 149–156.
Fontaine, K.M., Cooley, J.R., Simon, C., and Crusio, W. (2007). Evidence for paternal leakage in hybrid periodical cicadas (Hemiptera: *Magicicada* spp.). PLoS ONE 2, e892.
Forster, P., Harding, R., Torroni, A., and Bandelt, H.J. (1996). Origin and evolution of Native American mtDNA variation: a reappraisal. Am J Hum Genet 59, 935–945.
Forster, P (2004). Ice Ages and the mitochondrial DNA chronology of human dispersals: a review. Philos Trans R Soc Lond B Biol Sci 359, 1442 255–264.
Fuku, N., Park, K.S., Yamada, Y., Nishigaki, Y., Cho, Y.M., Matsuo, H., Segawa, T., Watanabe, S., Kato, K., Yokoi, K., et al. (2007). Mitochondrial haplogroup N9a confers resistance against Type 2 diabetes in Asians. Am J Hum Genet 80, 407–415.
Ghezzi, D., Marelli, C., Achilli, A., Goldwurm, S., Pezzoli, G., Barone, P., Pellecchia, M. T., Stanzione, P., Brusa, L., Bentivoglio, A.R., et al. (2005). Mitochondrial DNA haplogroup K is associated with a lower risk of Parkinson's disease in Italians. Eur J Hum Genet 13, 748–752.
Giles, R.E., Blanc, H., Cann, H.M., and Wallace, D.C. (1980). Maternal inheritance of human mitochondrial DNA.. Proc Natl Acad Sci USA 77, 6715–6719.
Glynos, A., Bozhilova, L.V., Frison, M., Burr, S., Stewart, J.B., and Chinnery, P.F. (2023). High-throughput single-cell analysis reveals progressive mitochondrial DNA mosaicism throughout life. Sci Adv 9, eadi4038.
González, M.M., Ramos, A., Aluja, M.P., and Santos, C. (2020). Sensitivity of mitochondrial DNA heteroplasmy detection using Next Generation Sequencing. Mitochondrion 50, 88–93.
Gopal, R.K., Calvo, S.E., Shih, A.R., Chaves, F.L., McGuone, D., Mick, E., Pierce, K.A., Li, Y., Garofalo, A., Van Allen, E.M., et al. (2018). Early loss of mitochondrial complex I and rewiring of glutathione metabolism in renal oncocytoma. Proc Natl Acad Sci USA 115, E6283–E6290.
Göransson, M., Andersson, M.K., Forni, C., Ståhlberg, A., Andersson, C., Olofsson, A., Mantovani, R., and Åman, P. (2009). The myxoid liposarcoma FUS-DDIT3 fusion oncoprotein deregulates NF-κB target genes by interaction with NFKBIZ. Oncogene 28, 270–278.
Gould, M.P., Bosworth, C.M., McMahon, S., Grandhi, S., Grimerg, B.T., LaFramboise, T., and Mishmar, D. (2015). PCR-free enrichment of mitochondrial DNA from human blood and cell lines for high quality next-generation DNA sequencing. PLoS ONE 10, e0139253.
Gunnarsdóttir, E.D., Nandineni, M.R., Li, M., Myles, S., Gil, D., Pakendorf, B., and Stoneking, M. (2011). Larger mitochondrial DNA than Y-chromosome differences between matrilocal and patrilocal groups from Sumatra. Nat Commun 2, 228.
Guo, Y., Li, C.I., Sheng, Q., Winther, J.F., Cai, Q., Boice, J.D., and Shyr, Y. (2013a). Very low-level heteroplasmy mtDNA variations are inherited in humans. J Genet Genomics 40, 607–615.
Guo, Y., Li, J., Li, C.I., Shyr, Y., and Samuels, D.C. (2013b). MitoSeek: extracting mitochondria information and performing high-throughput mitochondria sequencing analysis. Bioinformatics 29, 1210–1211.
Gupta, R., Kanai, M., Durham, T.J., Tsuo, K., McCoy, J.G., Kotrys, A.V., Zhou, W., Chinnery, P.F., Karczewski, K.J., Calvo, S.E., et al. (2023). Nuclear genetic control of mtDNA copy number and heteroplasmy in humans. Nature 620, 839–848.
Gyllensten, U., Wharton, D., Josefsson, A., and Wilson, A.C. (1991). Paternal inheritance of mitochondrial DNA in mice. Nature 352, 255–257.
Haber, M., Mezzavilla, M., Xue, Y., and Tyler-Smith, C. (2016). Ancient DNA and the rewriting of human history: be sparing with Occam's razor. Genome Biol 17, 1.
Hartman, P., Ponder, R., Lo, H.H., and Ishii, N. (2004). Mitochondrial oxidative stress can lead to nuclear hypermutability. Mech Ageing Dev 125, 417–420.
Hazkani-Covo, E., Zeller, R.M., Martin, W., and Malik, H.S. (2010). Molecular poltergeists: mitochondrial DNA copies (numts) in sequenced nuclear genomes. PLoS Genet 6, e1000834.
He, G., Wang, M., Luo, L., Sun, Q., Yuan, H., Lv, H., Feng, Y., Liu, X., Cheng, J., Bu, F., et al. (2024). Population genomics of Central Asian peoples unveil ancient Trans-Eurasian genetic admixture and cultural exchanges. hLife https://doi.org/10.1016/

j.hlife.2024.06.006.
He, Y., Wu, J., Dressman, D.C., Iacobuzio-Donahue, C., Markowitz, S.D., Velculescu, V. E., Diaz Luis A., J., Kinzler, K.W., Vogelstein, B., and Papadopoulos, N. (2010). Heteroplasmic mitochondrial DNA mutations in normal and tumour cells. Nature 464, 610–614.
Hoeh, W.R., Blakley, K.H., and Brown, W.M. (1991). Heteroplasmy suggests limited biparental inheritance of *Mytilus* mitochondrial DNA. Science 251, 1488–1490.
Holzinger, A., Langs, G., Denk, H., Zatloukal, K., and Müller, H. (2019). Causability and explainability of artificial intelligence in medicine. WIREs Data Min Knowl 9, e1312.
Howell, N., Smejkal, C.B., Mackey, D.A., Chinnery, P.F., Turnbull, D.M., and Herrnstadt, C. (2003). The pedigree rate of sequence divergence in the human mitochondrial genome: there is a difference between phylogenetic and pedigree rates. Am J Hum Genet 72, 659–670.
Hu, Z., Yang, L., Zhang, M., Tang, H., Huang, Y., Su, Y., Ding, Y., Li, C., Wang, M., Zhou, Y., et al. (2024). A novel protein CYTB-187AA encoded by the mitochondrial gene CYTB modulates mammalian early development. Cell Metab 36, 1586–1597.e7.
Hudson, G., Amati-Bonneau, P., Blakely, E.L., Stewart, J.D., He, L., Schaefer, A.M., Griffiths, P.G., Ahlqvist, K., Suomalainen, A., Reynier, P., et al. (2007). Mutation of OPA1 causes dominant optic atrophy with external ophthalmoplegia, ataxia, deafness and multiple mitochondrial DNA deletions: a novel disorder of mtDNA maintenance. Brain 131, 329–337.
Ji, F., Sharpley, M.S., Derbeneva, O., Alves, L.S., Qian, P., Wang, Y., Chalkia, D., Lvova, M., Xu, J., Yao, W., et al. (2012). Mitochondrial DNA variant associated with Leber hereditary optic neuropathy and high-altitude Tibetans. Proc Natl Acad Sci USA 109, 7391–7396.
Johns, D.R., Neufeld, M.J., and Park, R.D. (1992). An ND-6 mitochondrial DNA mutation associated with leber hereditary optic neuropathy. Biochem Biophys Res Commun 187, 1551–1557.
Jones, M.M. (2007). Mitochondrial DNA haplogroups and age-related maculopathy. Arch Ophthalmol 125, 1235–1240.
Ju, Y.S., Tubio, J.M.C., Mifsud, W., Fu, B., Davies, H.R., Ramakrishna, M., Li, Y., Yates, L., Gundem, G., Tarpey, P.S., et al. (2015). Frequent somatic transfer of mitochondrial DNA into the nuclear genome of human cancer cells. Genome Res 25, 814–824.
Kao, S.H., Chao, H.T., and Wei, Y.H. (1995). Mitochondrial deoxyribonucleic acid 4977-bp deletion is associated with diminished fertility and motility of human Sperm1. Biol Reprod 52, 729–736.
Katoh, K. (2002). MAFFT: a novel method for rapid multiple sequence alignment based on fast Fourier transform. Nucleic Acids Res 30, 3059–3066.
Kazuno, A., Munakata, K., Nagai, T., Shimozono, S., Tanaka, M., Yoneda, M., Kato, N., Miyawaki, A., Kato, T., and Orr, H. (2006). Identification of mitochondrial DNA polymorphisms that alter mitochondrial matrix pH and intracellular calcium dynamics. PLoS Genet 2, e128.
Keller, A., Graefen, A., Ball, M., Matzas, M., Boisguerin, V., Maixner, F., Leidinger, P., Backes, C., Khairat, R., Forster, M., et al. (2012). New insights into the Tyrolean Iceman's origin and phenotype as inferred by whole-genome sequencing. Nat Commun 3, 698.
Khusnutdinova, E., Gilyazova, I., Ruiz-Pesini, E., Derbeneva, O., Khusainova, R., Khidiyatova, I., Magzhanov, R., and Wallace, D.C. (2008). A mitochondrial etiology of neurodegenerative diseases: evidence from Parkinson's disease. Ann New York Acad Sci 1147, 1–20.
Kim, H.R., Won, S.J., Fabian, C., Kang, M.G., Szardenings, M., and Shin, M.G. (2015). Mitochondrial DNA aberrations and pathophysiological implications in hematopoietic diseases, chronic inflammatory diseases, and cancers. Ann Lab Med 35, 1–14.
Ko, A.M.S., Chen, C.Y., Fu, Q., Delfin, F., Li, M., Chiu, H.L., Stoneking, M., and Ko, Y.C. (2014). Early austronesians: into and out of Taiwan. Am J Hum Genet 94, 426–436.
Kogelnik, A. (1996). MITOMAP: a human mitochondrial genome database. Nucleic Acids Res 24, 177–179.
Kondo, R., Matsuura, E.T., and Chigusa, S.I. (1992). Further observation of paternal transmission of *Drosophila* mitochondrial DNA by PCR selective amplification method. Genet Res 59, 81–84.
Kong, Q.P., Bandelt, H.J., Sun, C., Yao, Y.G., Salas, A., Achilli, A., Wang, C.Y., Zhong, L., Zhu, C.L., Wu, S.F., et al. (2006). Updating the East Asian mtDNA phylogeny: a prerequisite for the identification of pathogenic mutations. Hum Mol Genet 15, 2076–2086.
Kong, Q.P., Sun, C., Wang, H.W., Zhao, M., Wang, W.Z., Zhong, L., Hao, X.D., Pan, H., Wang, S.Y., Cheng, Y.T., et al. (2011). Large-scale mtDNA Screening reveals a surprising matrilineal complexity in east Asia and its implications to the peopling of the region. Mol Biol Evol 28, 513–522.
Kurtz, S., Phillippy, A., Delcher, A.L., Smoot, M., Shumway, M., Antonescu, C., and Salzberg, S.L. (2004). Versatile and open software for comparing large genomes. Genome Biol 5, R12.
Kutanan, W., Kampuansai, J., Srikummool, M., Kangwanpong, D., Ghirotto, S., Brunelli, A., and Stoneking, M. (2017). Complete mitochondrial genomes of Thai and Lao populations indicate an ancient origin of Austroasiatic groups and demic diffusion in the spread of Tai–Kadai languages. Hum Genet 136, 85–98.
Laricchia, K.M., Lake, N.J., Watts, N.A., Shand, M., Haessly, A., Gauthier, L., Benjamin, D., Banks, E., Soto, J., Garimella, K., et al. (2022). Mitochondrial DNA variation across 56,434 individuals in gnomAD. Genome Res 32, 569–582.
Lazaridis, I., Patterson, N., Mittnik, A., Renaud, G., Mallick, S., Kirsanow, K., Sudmant, P.H., Schraiber, J.G., Castellano, S., Lipson, M., et al. (2014). Ancient human genomes suggest three ancestral populations for present-day Europeans. Nature 513, 409–413.
LeCun, Y., Bengio, Y., and Hinton, G. (2015). Deep learning. Nature 521, 436–444.
Lee, H.C., and Wei, Y.H. (1997). Mutation and oxidative damage of mitochondrial DNA and defective turnover of mitochondria in human aging. J Formos Med Assoc 96, 770–778.
Leonard, W.R., Sorensen, M.V., Galloway, V.A., Spencer, G.J., Mosher, M.J., Osipova, L., and Spitsyn, V.A. (2002). Climatic influences on basal metabolic rates among circumpolar populations. Am J Hum Biol 14, 609–620.
Li, W., Freudenberg, J., and Freudenberg, J. (2019a). Alignment-free approaches for predicting novel Nuclear Mitochondrial Segments (NUMTs) in the human genome. Gene 691, 141–152.
Li, F., Dong, S., Leier, A., Han, M., Guo, X., Xu, J., Wang, X., Pan, S., Jia, C., Zhang, Y., et al. (2022). Positive-unlabeled learning in bioinformatics and computational biology: a brief review. Brief BioInf 23, bbab461.
Li, M., Schönberg, A., Schaefer, M., Schroeder, R., Nasidze, I., and Stoneking, M. (2010). Detecting heteroplasmy from high-throughput sequencing of complete human mitochondrial DNA genomes. Am J Hum Genet 87, 237–249.
Li, M., Schröder, R., Ni, S., Madea, B., and Stoneking, M. (2015a). Extensive tissue-related and allele-related mtDNA heteroplasmy suggests positive selection for somatic mutations. Proc Natl Acad Sci USA 112, 2491–2496.
Li, M., Schroeder, R., Ko, A., and Stoneking, M. (2012). Fidelity of capture-enrichment for mtDNA genome sequencing: influence of NUMTs. Nucleic Acids Res 40, e137.
Li, X., Wang, M., Su, H., Duan, S., Sun, Y., Chen, H., Wang, Z., Sun, Q., Yang, Q., Chen, J., et al. (2024). Evolutionary history and biological adaptation of Han Chinese people on the Mongolian Plateau. hLife 2, 296–313.
Li, Y.C., Gao, Z.L., Liu, K.J., Tian, J.Y., Yang, B.Y., Rahman, Z.U., Yang, L.Q., Zhang, S. H., Li, C.T., Achilli, A., et al. (2023). Mitogenome evidence shows two radiation events and dispersals of matrilineal ancestry from northern coastal China to the Americas and Japan. Cell Rep 42, 112413.
Li, Y.C., Tian, J.Y., Liu, F.W., Yang, B.Y., Gu, K.S.Y., Rahman, Z.U., Yang, L.Q., Chen, F.H., Dong, G.H., and Kong, Q.P. (2019b). Neolithic millet farmers contributed to the permanent settlement of the Tibetan Plateau by adopting barley agriculture. Natl Sci Rev 6, 1005–1013.
Li, Y.C., Wang, H.W., Tian, J.Y., Liu, L.N., Yang, L.Q., Zhu, C.L., Wu, S.F., Kong, Q.P., and Zhang, Y.P. (2015b). Ancient inland human dispersals from Myanmar into interior East Asia since the Late Pleistocene. Sci Rep 5, 9473.
Libbrecht, M.W., and Noble, W.S. (2015). Machine learning applications in genetics and genomics. Nat Rev Genet 16, 321–332.
Lincoln, P., and Carracedo, A. (2000). Publication of population data of human polymorphisms. Forensic Sci Int 110, 3–5.
Lopez, J.V., Yuhki, N., Masuda, R., Modi, W., and O'Brien, S.J. (1994). Numt, a recent transfer and tandem amplification of mitochondrial DNA to the nuclear genome of the domestic cat. J Mol Evol 39, 174–190.
Luft, R., Ikkos, D., Palmieri, G., Ernster, L., and Afzelius, B. (1962). A case of severe hypermetabolism of nonthyroid origin with a defect in the maintenance of mitochondrial respiratory control: a correlated clinical, biochemical, and morphological study. J Clin Invest 41, 1776–1804.
Luo, S., Valencia, C.A., Zhang, J., Lee, N.C., Slone, J., Gui, B., Wang, X., Li, Z., Dell, S., Brown, J., et al. (2018). Biparental inheritance of mitochondrial DNA in humans. Proc Natl Acad Sci USA 115, 13039–13044.
Macaulay, V., Hill, C., Achilli, A., Rengo, C., Clarke, D., Meehan, W., Blackburn, J., Semino, O., Scozzari, R., Cruciani, F., et al. (2005). Single, rapid coastal settlement of Asia revealed by analysis of complete mitochondrial genomes. Science 308, 1034–1036.
Mardis, E.R. (2008). The impact of next-generation sequencing technology on genetics. Trends Genet 24, 133–141.
McArthur, K., Whitehead, L.W., Heddleston, J.M., Li, L., Padman, B.S., Oorschot, V., Geoghegan, N.D., Chappaz, S., Davidson, S., San Chin, H., et al. (2018). BAK/BAX macropores facilitate mitochondrial herniation and mtDNA efflux during apoptosis. Science 359, eaao6047.
Metzker, M.L. (2009). Sequencing technologies — the next generation. Nat Rev Genet 11, 31–46.

Meusel, M.S., and Moritz, R.F.A. (1993). Transfer of paternal mitochondrial DNA during fertilization of honeybee (*Apis mellifera* L.) eggs. Curr Genet 24, 539–543.
Michikawa, Y., Mazzucchelli, F., Bresolin, N., Scarlato, G., and Attardi, G. (1999). Aging-dependent large accumulation of point mutations in the human mtDNA control region for replication. Science 286, 774–779.
Min, S., Lee, B., and Yoon, S. (2016). Deep learning in bioinformatics. Brief Bioinform Bioinformatics, bbw068.
Mishmar, D., Ruiz-Pesini, E., Golik, P., Macaulay, V., Clark, A.G., Hosseini, S., Brandon, M., Easley, K., Chen, E., Brown, M.D., et al. (2002). Natural selection shaped regional mtDNA variation in humans. Proc Natl Acad Sci USA 100, 171–176.
Nandakumar, P., Tian, C., O'Connell, J., Hinds, D., Paterson, A.D., and Sondheimer, N. (2021). Nuclear genome-wide associations with mitochondrial heteroplasmy. Sci Adv 7, eabe7520.
Nishigaki, Y., Yamada, Y., Fuku, N., Matsuo, H., Segawa, T., Watanabe, S., Kato, K., Yokoi, K., Yamaguchi, S., Nozawa, Y., et al. (2007). Mitochondrial haplogroup N9b is protective against myocardial infarction in Japanese males. Hum Genet 120, 827–836.
Nomiyama, H., Fukuda, M., Wakasugi, S., Tsuzuki, T., and Shimada, K. (1985). Molecular structures of mitochondrial-DNA-Like sequences in human nuclear DNA. Nucleic Acids Res 13, 1649–1658.
Pääbo, S., Poinar, H., Serre, D., Jaenicke-Després, V., Hebler, J., Rohland, N., Kuch, M., Krause, J., Vigilant, L., and Hofreiter, M. (2004). Genetic analyses from ancient DNA. Annu Rev Genet 38, 645–679.
Pakendorf, B., and Stoneking, M. (2005). Mitochondrial DNA and human evolution. Annu Rev Genom Hum Genet 6, 165–183.
Parr, R.L., Maki, J., Reguly, B., Dakubo, G.D., Aguirre, A., Wittock, R., Robinson, K., Jakupciak, J.P., and Thayer, R.E. (2006). The pseudo-mitochondrial genome influences mistakes in heteroplasmy interpretation. BMC Genomics 7, 185.
Parson, W., and Dür, A. (2007). EMPOP—A forensic mtDNA database. Forensic Sci Int-Genet 1, 88–92.
Payne, B.A.I., and Chinnery, P.F. (2015). Mitochondrial dysfunction in aging: much progress but many unresolved questions. Biochim Biophys Acta (BBA)-Bioenergetics 1847, 1347–1353.
Peng, M.S., Palanichamy, M.G., Yao, Y.G., Mitra, B., Cheng, Y.T., Zhao, M., Liu, J., Wang, H.W., Pan, H., Wang, W.Z., et al. (2011). Inland post-glacial dispersal in East Asia revealed by mitochondrial haplogroup M9a'b. BMC Biol 9, 2.
Pereira, L., Soares, P., Radivojac, P., Li, B., and Samuels, D.C. (2011). Comparing phylogeny and the predicted pathogenicity of protein variations reveals equal purifying selection across the global human mtDNA diversity. Am J Hum Genet 88, 433–439.
Pinto, M., and Moraes, C.T. (2015). Mechanisms linking mtDNA damage and aging. Free Radical Biol Med 85, 250–258.
Poplin, R., Varadarajan, A.V., Blumer, K., Liu, Y., McConnell, M.V., Corrado, G.S., Peng, L., and Webster, D.R. (2018). Prediction of cardiovascular risk factors from retinal fundus photographs via deep learning. Nat Biomed Eng 2, 158–164.
Preste, R., Vitale, O., Clima, R., Gasparre, G., and Attimonelli, M. (2018). HmtVar: a new resource for human mitochondrial variations and pathogenicity data. Nucleic Acids Res 47, D1202–D1210.
Pyle, A., Hudson, G., Wilson, I.J., Coxhead, J., Smertenko, T., Herbert, M., Santibanez-Koref, M., Chinnery, P.F., and Larsson, N.G. (2015). Extreme-depth re-sequencing of mitochondrial DNA finds no evidence of paternal transmission in humans. PLoS Genet 11, e1005040.
Rahman, Z.U., Tian, J.Y., Gao, Z.L., Wang, H.T., Xia, W.X., Yang, B.Y., Yang, L.Q., Li, Y.C., and Kong, Q.P. (2021). Complete mitogenomes document substantial genetic contribution from the Eurasian Steppe into northern Pakistani Indo-Iranian speakers. Eur J Hum Genet 29, 1008–1018.
Ramos, A., Santos, C., Mateiu, L., Gonzalez, M.M., Alvarez, L., Azevedo, L., Amorim, A., Aluja, M.P., and Mishmar, D. (2013). Frequency and pattern of heteroplasmy in the complete human mitochondrial genome. PLoS ONE 8, e74636.
Rasmussen, M., Guo, X., Wang, Y., Lohmueller, K.E., Rasmussen, S., Albrechtsen, A., Skotte, L., Lindgreen, S., Metspalu, M., Jombart, T., et al. (2011). An aboriginal Australian genome reveals separate human dispersals into Asia. Science 334, 94–98.
Ratnaike, T.E., Greene, D., Wei, W., Sanchis-Juan, A., Schon, K.R., van den Ameele, J., Raymond, L., Horvath, R., Turro, E., and Chinnery, P.F. (2021). MitoPhen database: a human phenotype ontology-based approach to identify mitochondrial DNA diseases. Nucleic Acids Res 49, 9686–9695.
Rebolledo-Jaramillo, B., Su, M.S.W., Stoler, N., McElhoe, J.A., Dickins, B., Blankenberg, D., Korneliussen, T.S., Chiaromonte, F., Nielsen, R., Holland, M.M., et al. (2014). Maternal age effect and severe germ-line bottleneck in the inheritance of human mitochondrial DNA. Proc Natl Acad Sci USA 111, 15474–15479.
Rehm, H.L. (2017). Evolving health care through personal genomics. Nat Rev Genet 18, 259–267.
Ricchetti, M., Tekaia, F., Dujon, B., and Jonathan A. Eisen, B. (2004). Continued colonization of the human genome by mitochondrial DNA. PLoS Biol 2, e273.
Romesberg, A., and Van Houten, B. (2022). Targeting mitochondrial function with chemoptogenetics. Biomedicines 10, 2459.
Rossignol, R., Faustin, B., Rocher, C., Malgat, M., Mazat, J.P., and Letellier, T. (2003). Mitochondrial threshold effects. Biochem J 370, 751–762.
Ru, Y., Deng, X., Chen, J., Zhang, L., Xu, Z., Lv, Q., Long, S., Huang, Z., Kong, M., Guo, J., et al. (2024). Maternal age enhances purifying selection on pathogenic mutations in complex I genes of mammalian mtDNA. Nat Aging 4, 1211–1230.
Rubino, F., Piredda, R., Calabrese, F.M., Simone, D., Lang, M., Calabrese, C., Petruzzella, V., Tommaseo-Ponzetta, M., Gasparre, G., and Attimonelli, M. (2011). HmtDB, a genomic resource for mitochondrion-based human variability studies. Nucleic Acids Res 40, D1150–D1159.
Ruiz-Pesini, E., Mishmar, D., Brandon, M., Procaccio, V., and Wallace, D.C. (2004). Effects of purifying and adaptive selection on regional variation in human mtDNA. Science 303, 223–226.
Russell, S.J., and Norvig, P. (2010). Artificial Intelligence: A Modern Approach. London: Berkeley.
Sallevelt, S.C.E.H., de Die-Smulders, C.E.M., Hendrickx, A.T.M., Hellebrekers, D.M.E.I., de Coo, I.F.M., Alston, C.L., Knowles, C., Taylor, R.W., McFarland, R., and Smeets, H.J.M. (2017). *De novo* mtDNA point mutations are common and have a low recurrence risk. J Med Genet 54, 73–83.
Samuels, D.C., Wonnapinij, P., and Chinnery, P.F. (2013). Preventing the transmission of pathogenic mitochondrial DNA mutations: can we achieve long-term benefits from germ-line gene transfer? Hum Reprod 28, 554–559.
Sanchez-Contreras, M., Sweetwyne, M.T., Tsantilas, K.A., Whitson, J.A., Campbell, M. D., Kohrn, B.F., Kim, H.J., Hipp, M.J., Fredrickson, J., Nguyen, M.M., et al. (2023). The multi-tissue landscape of somatic mtDNA mutations indicates tissue-specific accumulation and removal in aging. eLife 12, e83395.
Schadt, E.E., Turner, S., and Kasarskis, A. (2010). A window into third-generation sequencing. Hum Mol Genet 19, R227–R240.
Schapira, A.H. (2006). Mitochondrial disease. Lancet 368, 70–82.
Schwartz, M., and Vissing, J. (2002). Paternal inheritance of mitochondrial DNA. N Engl J Med 347, 576–580.
Schwartz, M., and Vissing, J. (2004). No evidence for paternal inheritance of mtDNA in patients with sporadic mtDNA mutations. J Neurol Sci 218, 99–101.
Shen, L., Diroma, M.A., Gonzalez, M., Navarro-Gomez, D., Leipzig, J., Lott, M.T., van Oven, M., Wallace, D.C., Muraresku, C.C., Zolkipli-Cunningham, Z., et al. (2016). MSeqDR: a centralized knowledge repository and bioinformatics web resource to facilitate genomic investigations in mitochondrial disease. Hum Mutat 37, 540–548.
Sievers, F., and Higgins, D.G. (2017). Clustal Omega for making accurate alignments of many protein sequences. Protein Sci 27, 135–145.
Sirugo, G., Williams, S.M., and Tishkoff, S.A. (2019). The missing diversity in human genetic studies. Cell 177, 26–31.
Skoglund, P., Malmström, H., Raghavan, M., Storå, J., Hall, P., Willerslev, E., Gilbert, M.T.P., Götherström, A., and Jakobsson, M. (2012). Origins and genetic legacy of neolithic farmers and hunter-gatherers in Europe. Science 336, 466–469.
Snodgrass, J.J., Leonard, W.R., Sorensen, M.V., Tarskaia, L.A., and Mosher, M.J. (2008). The influence of basal metabolic rate on blood pressure among indigenous Siberians. Am J Phys Anthropol 137, 145–155.
Snodgrass, J.J., Leonard, W.R., Tarskaia, L.A., Alekseev, V.P., and Krivoshapkin, V.G. (2005). Basal metabolic rate in the Yakut (Sakha) of Siberia. Am J Hum Biol 17, 155–172.
Sondheimer, N., Glatz, C.E., Tirone, J.E., Deardorff, M.A., Krieger, A.M., and Hakonarson, H. (2011). Neutral mitochondrial heteroplasmy and the influence of aging. Hum Mol Genet 20, 1653–1659.
Steinborn, R., Zakhartchenko, V., Jelyazkov, J., Klein, D., Wolf, E., Müller, M., and Brem, G. (1998). Composition of parental mitochondrial DNA in cloned bovine embryos. FEBS Lett 426, 352–356.
Stewart, J.B., Freyer, C., Elson, J.L., Wredenberg, A., Cansu, Z., Trifunovic, A., Larsson, N.G., and Hurst, L.D. (2008). Strong purifying selection in transmission of mammalian mitochondrial DNA. PLoS Biol 6, e10.
Sugiyama, S., Hattori, K., Hayakawa, M., and Ozawa, T. (1991). Quantitative analysis of age-associated accumulation of mitochondrial DNA with deletion in human hearts. Biochem Biophys Res Commun 180, 894–899.
Sun, Y., Wang, M., Sun, Q., Liu, Y., Duan, S., Wang, Z., Zhou, Y., Zhong, J., Huang, Y., Huang, X., et al. (2024). Distinguished biological adaptation architecture aggravated population differentiation of Tibeto-Burman-speaking people. J Genet Genomics 51, 517–530.
Suomalainen, A., and Kaukonen, J. (2001). Diseases caused by nuclear genes affecting mtDNA stability. Am J Med Genet 106, 53–61.
Tang, S., and Huang, T. (2010). Characterization of mitochondrial DNA heteroplasmy using a parallel sequencing system. BioTechniques 48, 287–296.

Taylor, R.W., and Turnbull, D.M. (2005). Mitochondrial DNA mutations in human disease. Nat Rev Genet 6, 389–402.
Thompson, J.D., Higgins, D.G., and Gibson, T.J. (1994). CLUSTAL W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice. Nucleic Acids Res 22, 4673–4680.
Topol, E.J. (2019). High-performance medicine: the convergence of human and artificial intelligence. Nat Med 25, 44–56.
Torroni, A., Petrozzi, M., D'Urbano, L., Sellitto, D., Zeviani, M., Carrara, F., Carducci, C., Leuzzi, V., Carelli, V., Barboni, P., et al. (1997). Haplotype and phylogenetic analyses suggest that one European-specific mtDNA background plays a role in the expression of Leber hereditary optic neuropathy by increasing the penetrance of the primary mutations 11778 and 14484. Am J Hum Genet 60, 1107–1121.
Udar, N., Atilano, S.R., Memarzadeh, M., Boyer, D.S., Chwa, M., Lu, S., Maguen, B., Langberg, J., Coskun, P., Wallace, D.C., et al. (2009). Mitochondrial DNA haplogroups associated with age-related macular degeneration. Invest Ophthalmol Vis Sci 50, 2966–2974.
Uvizl, M., Puechmaille, S.J., Power, S., Pippel, M., Carthy, S., Haerty, W., Myers, E.W., Teeling, E.C., Huang, Z., and Bonatto, S. (2024). Comparative genome microsynteny illuminates the fast evolution of Nuclear Mitochondrial Segments (NUMTs) in mammals. Mol Biol Evol 41, msad278.
van der Walt, J.M., Dementieva, Y.A., Martin, E.R., Scott, W.K., Nicodemus, K.K., Kroner, C.C., Welsh-Bohmer, K.A., Saunders, A.M., Roses, A.D., Small, G.W., et al. (2004). Analysis of European mitochondrial haplogroups with Alzheimer disease risk. Neurosci Lett 365, 28–32.
van der Walt, J.M., Nicodemus, K.K., Martin, E.R., Scott, W.K., Nance, M.A., Watts, R. L., Hubble, J.P., Haines, J.L., Koller, W.C., Lyons, K., et al. (2003). Mitochondrial polymorphisms significantly reduce the risk of Parkinson disease. Am J Hum Genet 72, 804–811.
van Soest, D.M.K., Polderman, P.E., den Toom, W.T.F., Keijer, J.P., van Roosmalen, M. J., Leyten, T.M.F., Lehmann, J., Zwakenberg, S., De Henau, S., van Boxtel, R., et al. (2024). Mitochondrial H2O2 release does not directly cause damage to chromosomal DNA. Nat Commun 15, 2725.
Vellarikkal, S.K., Dhiman, H., Joshi, K., Hasija, Y., Sivasubbu, S., and Scaria, V. (2015). mit-o-matic: a comprehensive computational pipeline for clinical evaluation of mitochondrial variations from next-generation sequencing datasets. Hum Mutat 36, 419–424.
Vernot, B., Zavala, E.I., Gómez-Olivencia, A., Jacobs, Z., Slon, V., Mafessoni, F., Romagné, F., Pearson, A., Petr, M., Sala, N., et al. (2021). Unearthing Neanderthal population history using nuclear and mitochondrial DNA from cave sediments. Science 372, eabf1667.
Vieira, S., Lopez Pinaya, W.H., and Mechelli, A. (2020). Introduction to machine learning. In: Mechelli A., and Vieira S., eds. Machine Learning: Methods and Applications to Brain Disorders. New York: Academic Press 1–20.
Wallace, D.C. (1999). Mitochondrial diseases in man and mouse. Science 283, 1482–1488.
Wallace, D.C. (2005). A mitochondrial paradigm of metabolic and degenerative diseases, aging, and cancer: a dawn for evolutionary medicine. Annu Rev Genet 39, 359–407.
Wallace, D.C., Ye, J., Neckelmann, S.N., Singh, G., Webster, K.A., and Greenberg, B.D. (1987). Sequence analysis of cDNAs for the human and bovine ATP synthase? subunit: mitochondrial DNA genes sustain seventeen times more mutations. Curr Genet 12, 81–90.
Wallace, D.C., Zheng, X., Lott, M.T., Shoffner, J.M., Hodge, J.A., Kelley, R.I., Epstein, C. M., and Hopkins, L.C. (1988). Familial mitochondrial encephalomyopathy (MERRF): genetic, pathophysiological, and biochemical characterization of a mitochondrial DNA disease. Cell 55, 601–610.
Walsh, D.J., Bernard, D.J., Pangilinan, F., Esposito, M., Harold, D., Parle-McDermott, A., and Brody, L.C. (2022). Mito-SiPE is a sequence-independent and PCR-free mtDNA enrichment method for accurate ultra-deep mitochondrial sequencing. Commun Biol 5, 1269.
Wanagat, J., Cao, Z., Pathare, P., and Aiken, J.M. (2001). Mitochondrial DNA deletion mutations colocalize with segmental electron transport system abnormalities, muscle fiber atrophy, fiber splitting, and oxidative damage in sarcopenia. FASEB Journal 15, 322–332.
Wang, M., Chen, H., Luo, L., Huang, Y., Duan, S., Yuan, H., Tang, R., Liu, C., and He, G. (2024a). Forensic investigative genetic genealogy: expanding pedigree tracing and genetic inquiry in the genomic era. J Genet Genomics S1673-8527(24)00158-9.
Wang, M., Huang, Y., Liu, K., Wang, Z., Zhang, M., Yuan, H., Duan, S., Wei, L., Yao, H., Sun, Q., et al. (2024b). Multiple human population movements and cultural dispersal events shaped the landscape of chinese paternal heritage. Mol Biol Evol 41, msae122.
Wang, T., Antonacci-Fulton, L., Howe, K., Lawson, H.A., Lucas, J.K., Phillippy, A.M., Popejoy, A.B., Asri, M., Carson, C., Chaisson, M.J.P., et al. (2022). The Human Pangenome Project: a global resource to map genomic diversity. Nature 604, 437–446.
Wei, W., Pagnamenta, A.T., Gleadall, N., Sanchis-Juan, A., Stephens, J., Broxholme, J., Tuna, S., Odhams, C.A., Ambrose, J.C., Baple, E.L., et al. (2020). Nuclear-mitochondrial DNA segments resemble paternally inherited mitochondrial DNA in humans. Nat Commun 11, 1740.
Wei, W., Schon, K.R., Elgar, G., Orioli, A., Tanguy, M., Giess, A., Tischkowitz, M., Caulfield, M.J., and Chinnery, P.F. (2022). Nuclear-embedded mitochondrial DNA sequences in 66,083 human genomes. Nature 611, 105–114.
Weissensteiner, H., Forer, L., Fuchsberger, C., Schöpf, B., Kloss-Brandstätter, A., Specht, G., Kronenberg, F., and Schönherr, S. (2016). mtDNA-Server: next-generation sequencing data analysis of human mitochondrial DNA in the cloud. Nucleic Acids Res 44, W64–W69.
West, A.P., and Shadel, G.S. (2017). Mitochondrial DNA in innate immune responses and inflammatory pathology. Nat Rev Immunol 17, 363–375.
Willis, K.J., Bennett, K.D., Walker, D., and Forster, P. (2004). Ice Ages and the mitochondrial DNA chronology of human dispersals: a review. Phil Trans R Soc Lond B 359, 255–264.discussion 264.
Wittig, H., Koecke, M., Sattler, K.U., and Krause, D. (2003). D-Loop-BASE is online now Central European database of mitochondrial DNA. Int Congress Ser 1239, 505–509.
Wolff, J.N., Nafisinia, M., Sutovsky, P., and Ballard, J.W.O. (2012). Paternal transmission of mitochondrial DNA as an integral part of mitochondrial inheritance in metapopulations of Drosophila simulans. Heredity 110, 57–62.
Yang, L., Lin, X., Tang, H., Fan, Y., Zeng, S., Jia, L., Li, Y., Shi, Y., He, S., Wang, H., et al. (2020). Mitochondrial DNA mutation exacerbates female reproductive aging via impairment of the NADH/NAD$^+$ redox. Aging Cell 19, e13206.
Yang, L., Ruan, Z., Lin, X., Wang, H., Xin, Y., Tang, H., Hu, Z., Zhou, Y., Wu, Y., Wang, J., et al. (2024). NAD+ dependent UPRmt activation underlies intestinal aging caused by mitochondrial DNA mutations. Nat Commun 15, 546.
Yao, Y.G., Kong, Q.P., Bandelt, H.J., Kivisild, T., and Zhang, Y.P. (2002). Phylogeographic differentiation of mitochondrial DNA in Han Chinese. Am J Hum Genet 70, 635–651.
Zaidi, A.A., Wilton, P.R., Su, M.S.W., Paul, I.M., Arbeithuber, B., Anthony, K., Nekrutenko, A., Nielsen, R., and Makova, K.D. (2019). Bottleneck and selection in the germline and maternal age influence transmission of mitochondrial DNA in human pedigrees. Proc Natl Acad Sci USA 116, 25172–25178.
Zaragoza, M.V., Fass, J., Diegoli, M., Lin, D., Arbustini, E., and Li, W. (2010). Mitochondrial DNA variant discovery and evaluation in human cardiomyopathies through next-generation sequencing. PLoS ONE 5, e12295.
Zhang, D., Xia, H., Chen, F., Li, B., Slon, V., Cheng, T., Yang, R., Jacobs, Z., Dai, Q., Massilani, D., et al. (2020). Denisovan DNA in Late Pleistocene sediments from Baishiya Karst Cave on the Tibetan Plateau. Science 370, 584–587.
Zhao, M., Kong, Q.P., Wang, H.W., Peng, M.S., Xie, X.D., Wang, W.Z., Jiayang, W.Z., Duan, J.G., Cai, M.C., Zhao, S.N., et al. (2009). Mitochondrial genome evidence reveals successful Late Paleolithic settlement on the Tibetan Plateau. Proc Natl Acad Sci USA 106, 21230–21235.
Zheng, H.X., Yan, S., Qin, Z.D., Wang, Y., Tan, J.Z., Li, H., Jin, L., and Kivisild, T. (2011). Major population expansion of east Asians began before neolithic time: evidence of mtDNA genomes. PLoS ONE 6, e25835.
Zhidkov, I., Nagar, T., Mishmar, D., and Rubin, E. (2011). MitoBamAnnotator: a web-based tool for detecting and annotating heteroplasmy in human mitochondrial DNA sequences. Mitochondrion 11, 924–928.
Zou, J., Huss, M., Abid, A., Mohammadi, P., Torkamani, A., and Telenti, A. (2018). A primer on deep learning in genomics. Nat Genet 51, 12–18.